Journal of Banking and Finance 163 (2024) 107184

Contents lists available at ScienceDirect

Journal of Banking and Finance

journal homepage: www.elsevier.com/locate/jbf

Limit-hitting exciting effects: Modeling jump dependencies in stock markets adhering to daily price-limit rules ☆

Jian Chen [a], Shuyuan Qi [b,*]

[a] Business School, University of Sussex, Brighton, BN1 9SL, United Kingdom
[b] Chinese Academy of Finance and Development, Central University of Finance and Economics, Beijing, 100081, China

ARTICLE INFO

JEL classification:
C11
C58
G12

Keywords:
Price limits
Jump clustering
Bayesian inference
Hawkes process
Stochastic volatility

ABSTRACT

Price limits are widely implemented in stock markets worldwide; however, they are rarely considered in financial models. In this study, we propose a model specifically designed for asset prices that adhere to daily price-limit mechanisms. Our model captures the interdependence among limit-hitting events and other small price jumps by using a multivariate mutually-exciting point process. It is applicable to any stock market with a multi-layer price limit mechanism. By analyzing data from all publicly listed A-share stocks in China from 2007 to 2021, we demonstrate that our model outperforms other classic models in terms of goodness of fit. Additionally, we find that limit-hitting jumps, as opposed to inconspicuous small price jumps, have a higher propensity to attract investors' attention and result in subsequent price jumps. We further construct a clustering index based on the model parameters and investigate its determinants.

1. Introduction

The implementation of daily price-limit mechanisms is a prevalent market stabilization mechanism utilized in stock exchanges globally. Numerous countries, including China and Japan (the second and the third-largest stock market), adhere to this mechanism.[1] Hitting the price limits can potentially trigger subsequent turbulent market reactions. However, the interdependence between limit-hitting (LH) events and other jumps is rarely considered in traditional financial models. This study aims to address this gap by proposing a model specifically tailored for asset prices subject to daily price-limit mechanisms. The model is designed to capture the interdependence among limit-hitting events and other small price jumps using a multivariate mutually-exciting point process, and the new model offers a more accurate representation of market dynamics.

Specifically, the model distinguishes between limit-hitting jumps and other small jumps and further categorizes them into positive and negative jumps. We adopt a double exponential process, as suggested by Kou et al. (2017), to model the sizes of limit-hitting jumps. This choice is motivated by their argument that the sizes of large jumps are more

☆ Part of this collaborative work was accomplished while Jian Chen and Shuyuan Qi were pursuing their PhD at the ICMA Centre, University of Reading. We deeply appreciate the support received from the ICMA Centre. We are immensely grateful for the insightful feedback and guidance provided by the editor, associate editor, and two anonymous referees. Our heartfelt thanks also go to the seminar attendees at the University of Reading and the Central University of Finance and Economics for sharing their invaluable suggestions. Additionally, we would like to express our gratitude to the participants of the 20th Chinese Finance Annual Meeting, BAFA-SWAG in Bristol, the 8th China Accounting and Finance International Conference, the 5th International Conference on Financial Regulation and Financial Stability, and the 12th International Symposium of Quantitative Economics for their constructive feedback and suggestions.

* Corresponding author.
E-mail addresses: jian.chen@sussex.ac.uk (J. Chen), shuyuan.qi@cufe.edu.cn (S. Qi).

[1] Price limit mechanism typically sets predetermined thresholds that trigger trading restrictions. For example, if the threshold is set as 10% for a stock, the stock is not allowed to be traded outside the range of ±10% of the previous closing price. Furthermore, the list of countries by stock market capitalization can be easily found in any database. We retrieve this information from Bloomberg. In addition to China and Japan, many other countries, such as Austria, Belgium, France, Greece, Italy, Korea, Malaysia, Thailand, etc., also apply price limits in their stock markets. A more detailed summary of equity exchanges that apply price-limit mechanisms can be found in Deb et al. (2010).

https://doi.org/10.1016/j.jbankfin.2024.107184
Received 6 August 2023; Accepted 12 April 2024
Available online 21 April 2024
0378-4266/© 2024 Elsevier B.V. All rights reserved.

suitable to be modeled by an exponential distribution rather than a normal distribution.[2] Following the standard literature, we model the sizes of other small jumps with a normal distribution. Crucially, we incorporate a multivariate Hawkes process (MHP) to model different jump intensities, enabling us to capture the dependencies among jumps.

The MHP is proposed in Hawkes (1971a,b); it relaxes the independent assumption of events and has attracted applications in various financial contexts. For example, Aït-Sahalia et al. (2015) incorporate a mutually-exciting jump process into an Itô's semimartingale to study the cross-impact of international markets. Maneesoonthorn et al. (2017) detect asset price jumps using high-frequency measures and show the self-excitation behaviors of jumps. Du and Luo (2019) investigate the jump propagation phenomenon in the US market using a two-factor Hawkes jump-diffusion model.

The empirical studies in this paper are conducted using all the A-share stocks in the Chinese market, which is the largest market that applies price limit rules. We begin by assessing the performance of our proposed model against four benchmark models. Our model best fits the stock returns in terms of the deviance information criterion (DIC) and the log-likelihood (LLK). In addition, the results obtained from our model align with the empirical findings reported in the existing literature. For instance, we find that LH jumps can increase the probability of future jumps, while other inconspicuous small price jumps are less likely to have such effects, implying the existence of the limit-hitting-exciting (LHE) effects. This observation is consistent with the argument put forth by Seasholes and Wu (2007), who contend that LH jumps are more likely to attract investors' attention. Furthermore, we also find evidence of disruptive market behaviors (first discovered by Chen et al., 2019), in that LH jumps can increase the probability of jumps in the opposite directions.

To further illustrate the usefulness and importance of our model, we propose a clustering index, calculated by using model-estimated parameters, to measure the interdependence among LH and other jumps. Notably, the clustering index of LH jumps is found to be significantly higher than that of small jumps. To explore the implications of this clustering index, we conduct a portfolio analysis by sorting stocks based on their clustering index values. The results of our study reveal several key findings. First, stocks exhibiting more significant LHE effects are typically large-cap liquid stocks, characterized by high trading volumes and fat-tailed return distributions. By contrast, stocks exhibiting higher clustering effects of small jumps tend to be small-cap illiquid stocks with low trading volumes and thin-tailed return distributions. Furthermore, we investigate the factors associated with LHE effects and find that certain factors related to investor attention, for example, media coverage and institutional holdings, partially account for these effects. We also investigate the association between market sentiment and market-level LHE effects.

The rationale of the LHE effects incorporated in our model comes from three popular hypotheses regarding price limit critics documented by Kim and Rhee (1997); namely: delayed price discovery, volatility spillover, and trading interference. To begin, the delayed price discovery hypothesis, emphasized by Fama (1988) and Lehmann (1989), suggests that price limits impede the process of price discovery. Specifically, price-limit rules prevent stock prices from promptly reflecting market information and reaching their equilibrium levels. Consequently, in line with the efficient market hypothesis, stocks are expected to continue moving toward their "equilibrium" prices in the subsequent trading period, thereby increasing the propensity for further price jumps.

Additionally, the volatility spillover hypothesis states that the supply and demand of shares are unbalanced when hitting price limits, and trades that have not been executed will be pushed to the following days. This can lead to higher volatility with large price fluctuations, which is often accompanied by a higher probability of jumps (Santa-Clara and Yan, 2010; John and Li, 2021). Furthermore, Bates (2000) and Andersen et al. (2002) model the jump intensity as a function of the instantaneous volatility, which underpins the idea that LH events may lead to higher volatility with higher probabilities of jumps.

Furthermore, the trading interference hypothesis suggests that a surge in trading volume is likely to occur in the days following price limit hits. (Lauterbach and Ben-Zion, 1993). Lehmann (1989) also argues that order imbalances resulting from limit hits, coupled with the subsequent lack of trading, can drive prices toward their limit levels. Consequently, impatient investors may engage in trading at unfavorable prices, leading to increased trading volume. Such intensified trading activity could also lead to price jumps. Additionally, Seasholes and Wu (2007) show that LH events could attract the attention of investors in the following days, while Jiang et al. (2022) show that anomaly returns tend to be higher during high-attention days. Moreover, Chen et al. (2019) find that the trading activities after limit hits vary across different investor groups. They document disruptive trading behavior by large investors, who tend to push stocks to the upper price limit at market closes and subsequently profit from selling at higher prices the next day.

Given substantial evidence that limit-hitting events can increase the probability of subsequent price jumps in stock markets, which we refer to as the LHE effects in this study, it becomes evident that traditional jump-diffusion stochastic volatility models may yield biased results when applied in these markets. This bias arises from the underlying assumption of jump components in the model. The incorporation of jump components in stochastic financial models has a long history (see e.g., Merton, 1976; Bates, 1996; Duffie et al., 2000; Eraker, 2004). Most of the existing literature adopts the assumption that jumps are serially independent to ensure mathematical tractability. However, this assumption is no longer appropriate for the stock markets with price-limit rules due to the LHE effects.

Classic models not only fail to account for price limit rules, but also fail to capture the dynamics of the LHE effects. They do not account for the specific characteristics and interdependence associated with limit-hitting events and their impact on subsequent price movements. Therefore, there is a need for a model that explicitly considers price limit rules and incorporates the LHE effects to provide a more accurate representation of the market dynamics.

Overall, this study contributes to the literature by offering a model that explicitly considers price limit rules and incorporates the LHE effects, thereby enhancing our understanding of market dynamics within markets adhering to price-limit mechanisms. The rest of this paper is organized as follows. Section 2 introduces our model setup and estimation methods. Section 3 provides details on the data used in our empirical analysis and presents the corresponding empirical results. Section 4, 5 and 6 deliver the empirical results. Section 7 concludes the paper.

2. Model and estimation methods

This section introduces our model specifications, including the price/variance process and the multivariate Hawkes process (MHP). We also introduce the estimation methods in this section.

2.1. Continuous-time representation

We model the natural logarithm of an asset price p_t that evolves in continuous time via:

$$\begin{aligned} dp_t &= \mu dt + \sqrt{v_t} dW_t^p + \xi_t dN_t, \\ dv_t &= \kappa(\theta - v_t)dt + \sqrt{v_t}\sigma_v dW_t^v. \end{aligned} \tag{1}$$

[2] The size of limit-hitting jumps in this paper refers to the jump component of returns when the prices hit the limit. We remark that their sizes are not necessarily the boundaries of returns (e.g., ± 10%) since there is a Brownian motion and a drift in the return process. Therefore, it is imperative to model their sizes rather than to allow them to equal the boundary values.

Here, μ is the drift term; v_t represents the variance process; κ and θ denote the mean reversion rate and long-run mean of variance v_t, respectively; and σ_v represents the volatility of volatility. The terms dW_t^p and dW_t^v are two correlated Brownian motions with $corr(dW_t^p, dW_t^v) = \rho$; The term $\xi_t dN_t$ represents the price jump component. The specifications at this level resemble the stochastic volatility model with jumps (SVJ) proposed by Bates (1996). However, considering the price limit rules, we model the price jump component differently as follows:

$$\xi_t dN_t = \sum_i \xi_t^i dN_t^i, \qquad \mathbf{P}(dN_t^i = 1) = \lambda_t^i, \qquad i \in \{pl^+, pl^-, y^+, y^-\}. \quad (2)$$

Where λ_t^i denotes the underlying intensity of jumps in dimension i. We specify four different dimensions, $i \in \{pl^+, pl^-, y^+, y^-\}$, regarding upper and lower bound limit hits (pl^+ and pl^-), and other positive and negative small price jumps (y^+ and y^-), respectively. Small jumps refer to those price jumps that do not reach the price limits.

The above specification makes our model more applicable to markets with price limit rules. As discussed in Section 1, the dependencies among limit-hitting jumps and other small jumps are the key features of the financial market with price limit rules. To capture this interdependence, we model the jump intensity λ_t^i using a MHP:

$$\lambda_t^i = \lambda_0^i + \sum_j \int_{[0,t)} \alpha_{j,i} e^{-\beta_{j,i}(t-s)} N_j ds, \quad i, j \in \{pl^+, pl^-, y^+, y^-\}, \quad (3)$$

where $i, j \in \{pl^+, pl^-, y^+, y^-\}$; λ_0^i is a constant baseline intensity for jumps in the dimension i; $\alpha_{j,i}$ and $\beta_{j,i}$ are two parameters in the MHP; $\alpha_{j,i}$ captures the incremental intensity of jumps in dimension i given the occurrence of jumps in dimension j; while $\beta_{j,i}$ represents the decay rate of the incremental intensity.

For a specific example, considering $\lambda_t^{y^+}$, which represents the underlying intensity of positive jumps, and $\lambda_t^{y^+} = \lambda_0^{y^+}$ during peaceful times. However, when a jump occurs in the dimension pl^+ (indicating that the price hits the upper price limit), the underlying intensity of positive return jump y^+ will increase to $\lambda_{t+1}^{y^+} = \lambda_0^{y^+} + \alpha_{pl^+,y^+}$. The incremental intensity α_{pl^+,y^+} will decay over time at a speed of $e^{-\beta_{pl^+,y^+} ds}$ for each small time interval ds. Therefore, the underlying intensity λ_t^i is composed of a baseline intensity λ_0^i and incremental intensities produced by past jumps across all dimensions, resulting in a self/mutually-exciting effect known as jump clustering.

Regarding the jump size ξ_t^i, we model the LH jumps and other small jumps differently. We let LH jumps follow a double exponential distribution with mean $\{\mu_{pl^+}, \mu_{pl^-}\}$, and other small jumps follow a normal distribution[3]:

$$\begin{cases} \xi_t^{pl^+} \sim EXP(\mu_{pl^+}), \xi_t^{pl^-} \sim -EXP(\mu_{pl^-}), \\ \xi_t^{y^+}, \xi_t^{y^-} \sim \mathcal{N}(\mu_p, \sigma_p). \end{cases} \quad (4)$$

Note we let $\xi_t^{y^+}$ and $\xi_t^{y^-}$ jointly follow a normal distribution with common parameters μ_p and σ_p, and model their underlying intensities separately to capture the cross-impact. Furthermore, we categorize the LH jumps into positive and negative, which matches the feature of LH events where the price can either hit its upper or lower limit.

The motivation behind modeling LH jumps and other small jumps differently is based on the insights from Kou et al. (2017). The authors argue that the double-exponential distribution is better suited for modeling large jumps, which are characteristic of LH events, owing to its heavier tails compared to the normal distribution. LH events are more likely to result in large price jumps; thus, the jump sizes are more suitable to be described by the exponential distribution. Conversely, other small jumps are associated with a relatively smaller size, not big enough to hit the price limits. The normal distribution, with its relatively thinner tails compared to the exponential distribution, is more appropriate to capture the sizes of these small jumps.

For clarity, in the previous formulation, we only consider the case where the market applies only a single-layer price limit mechanism; however, multi-layer price limits are commonly implemented in many stock markets in the world. For example, China's equity market imposes daily price limits of 10% on regular stocks, 5% on special treatment (ST) stocks and 20% on listing companies on the ChiNext Board.[4] Our model can be expanded to accommodate multi-layer price limits by introducing different jump sizes for each price limit based on market regulations. Specifically, we denote the jump sizes as $\xi_t^{pl^+,m} \sim EXP(\mu_{pl^+}^m)$ and $\xi_t^{pl^-,m} \sim -EXP(\mu_{pl^-}^m)$, where m represents the different layers of price limits. By specifying appropriate jump size parameters for each price limit layer, our model can effectively capture the dynamics of multi-layer price limits in stock markets. For a more specific example, please see Section 3.2 and the third column graphs of Fig. 1, where we experiment with a stock having multi-layer price limits in different periods.

2.2. Discretized form

To estimate this model using daily data, we employ an Euler's discretization with a time step of $\Delta = \frac{1}{252}$ and obtain the following discretized forms:

$$p_t = p_{t-1} + \mu\Delta + \sqrt{v_t \Delta}\epsilon_t^p + \xi_t J_t, \quad (5)$$

$$v_t = v_{t-1} + \kappa(\theta - v_{t-1})\Delta + \sqrt{v_{t-1}\Delta}\sigma_v \epsilon_t^v. \quad (6)$$

Here, the parameters $\mu, \kappa, \theta, \sigma_v$ have the same interpretations as in the continuous-time representation (Section 2.1). The terms ϵ_t^p and ϵ_t^v are two random variables drawn from $\mathcal{N}(0,1)$, and $corr(\epsilon_t^p, \epsilon_t^v) = \rho$. The distribution of jump sizes, ξ_t, remains the same as in Eq. (4), and $J_t \sim Bernoulli(\lambda_t)$. Again, we consider four different dimensions $\{pl^+, pl^-, y^+, y^-\}$ for the jump components, and their underlying intensities are modeled using an MHP:

$$\lambda_t^i = \lambda_0^i + \sum_j \sum_{0<s<t} \alpha_{j,i} e^{-\beta_{j,i}(t-s)}, \quad i, j \in \{pl^+, pl^-, y^+, y^-\}. \quad (7)$$

Notably, if there are no jump clustering features, the jump component reduces to a Poisson process with $\lambda_t^i \equiv \lambda_0^i$ and $\alpha_{j,i} \equiv 0$. The presence of non-zero values for $\alpha_{j,i}$ indicates the existence of self-/mutual-exciting effects.

2.3. Model estimation

We estimate the model using the Markov Chain Monte Carlo (MCMC) approach.[5] We denote the static parameter vector as $\Theta = \{\mu, \kappa, \theta, \sigma_v, \rho, \mu_p, \sigma_p, \mu_{pl^+}, \mu_{pl^-}, \boldsymbol{\lambda_0}, \boldsymbol{\alpha}, \boldsymbol{\beta}\}$. Here, $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are two 4×4 matrices representing the parameters in the MHP. Since we have four dimensions $\{pl^+, pl^-, y^+, y^-\}$, $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ become 4×4 matrices

[3] EXP represents the exponential distribution; and $\mathcal{N}$ represents the normal distribution.

[4] An ST firm has certain abnormalities in its financial position or other elements that might make it challenging for investors to assess the company's prospects. There are separate daily price limit rules that exist for regular stocks and ST status stocks. Regular stocks are subject to a 10% daily price limit, whereas ST stocks have a 5% daily price limit in either direction.

[5] Compared to other classical estimation methods, e.g., the maximum-likelihood estimation method and the generalized method of moments, the MCMC methods provide many benefits that are particularly appropriate for our investigation. For example, the MCMC not only estimates the unknown parameters but also provides posterior estimated latent variables such as stochastic volatility, jump times, and jump sizes; these are all critical variables for further empirical analysis and model comparison. Additionally, the MCMC approach is extremely efficient in terms of implementation.

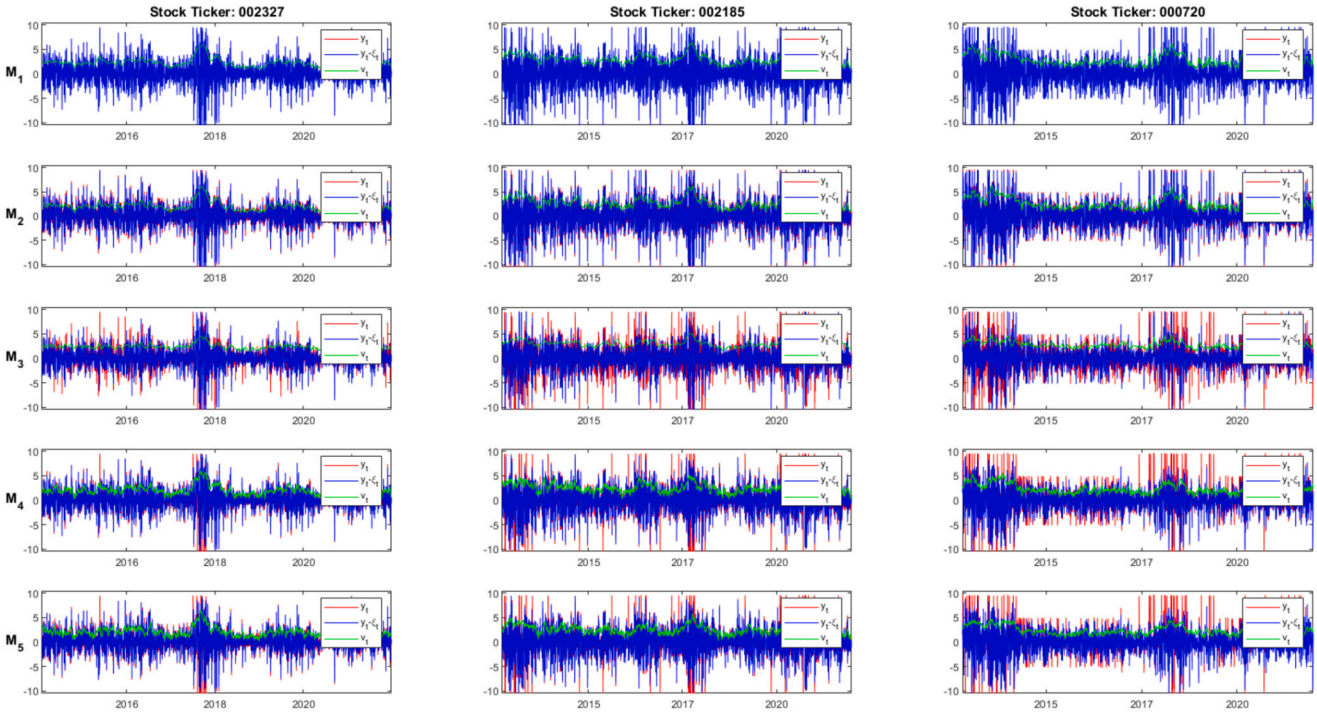

Fig. 1. Stock daily returns, returns minus estimated jumps and estimated volatility for the three A-share stocks. Jumps and volatility are estimated by different models listed in Table 2. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Table 1
Descriptive statistics of price limit hittings.

Variable	# of Stocks	Mean	Std. dev.	Min	Max
Total PLH	3646	94.14	64.58	4	301
PLH per year	3646	7.91(3.14%)	5.73(2.27%)	0.55(0.22%)	26.62(10.56%)

Note: This table reports the descriptive statistics of PLH events. Total PLH stands for total PLH events during the data period, while PLH per year is the PLH events observed per trading year. The percentage in the bracket of the last row stands for the frequency of PLH events.

and $\boldsymbol{\lambda_0}$ is a 4×1 vector corresponding to these four dimensions. We also denote the vector of latent variables as $\Upsilon_t = \{\upsilon_t, \Xi_t, \boldsymbol{J}_t\}$, where $\Xi_t = \{\xi_t^{pl^+}, \xi_t^{pl^-}, \xi_t^{y^+}, \xi_t^{y^-}\}$ and $\boldsymbol{J}_t = \{\boldsymbol{J}_t^{pl^+}, \boldsymbol{J}_t^{pl^-}, \boldsymbol{J}_t^{y^+}, \boldsymbol{J}_t^{y^-}\}$. We remark that LH jumps ($\{\boldsymbol{J}_t^{pl^+}, \boldsymbol{J}_t^{pl^-}\}$) are observable and do not require estimation, while their corresponding jump sizes do require estimation.

We conduct a Bayesian inference of static parameters and latent variables. The joint posterior distribution is given by:

$$\mathbf{P}\left(\Theta, \Upsilon_t | p_t\right) \propto \mathbf{P}\left(p_t | \Theta, \Upsilon_t\right) \mathbf{P}\left(\Upsilon_t | \Theta\right) \mathbf{P}(\Theta). \tag{8}$$

We adopt the MCMC methods to sample from the posterior distribution sequentially. We follow Lazar and Qi (2022) and Chen et al. (2023) to construct our prior settings and posterior distributions; a more detailed description can be found in the Appendix.

3. Empirical analysis

3.1. Data

We consider all publicly listed A-share stocks in China; the daily stock price data is collected from the China Stock Market and Accounting Research (CSMAR) database. We retrieve daily stock data from January 2007 to December 2021. We use the closing prices of each stock to identify PLH events (as is done by the stock exchanges and in Chen et al., 2019).

We filter our data by the following rules: 1) exclude listed companies that have not completed equity division reform,[6] we retain data only after the completion of this reform to ensure consistency; 2) exclude companies that are listed after January 2020 because smaller data size can result in less informative outcomes; 3) exclude 79 stocks that experience no LH events throughout the entire data period. Consequently, our database contains 3646 stocks; we then make the following further processing on the data: 1) delete data where closing prices are not within the range of daily high and low prices; 2) delete first-month data of every company listed after January 2007[7]; and 3) regarding stocks that experienced suspension and resumption, there are no trading price limits on the first trading days after their resumptions, and returns are typically extremely high or low reflecting information of corresponding companies during their suspensions; in this case, we define returns that exceed the price limits as hitting the price limits with corresponding limit sizes. Table 1 presents descriptive statistics of PLH events. Notably,

[6] Listing corporation equity division reform is implemented by China Securities Regulatory Commission (CSRC) from 2005. It aims to release trading restrictions of 'untradable' stocks of some previously listed companies. It was mostly completed by January 2007. In alignment with other data we collect, we exclude stock data that predates the completion of the equity division reform.

[7] CSRC imposed restrictions on the stock issue price, requiring associated P/E ratio to be lower than 23. This policy results in a high occurrence of continuous upper-price-limit-hitting events for newly listed stocks within their first month, which is not relevant to our research objective.

Table 2
Model specifications.

Model	Jump Specifications
$\mathcal{M}_1$ (Heston, 1993)	No jump components
$\mathcal{M}_2$ (Bates, 1996)	No distinction between LH jumps and small jumps; jump sizes are modeled by a Normal distribution; jump intensity is constant
$\mathcal{M}_3$ (Kou and Wang, 2004)	No distinction between LH jumps and small jumps; jump sizes are modeled by double exponential distributions; jump intensity is constant
$\mathcal{M}_4$ (NEW)	LH jump sizes are modeled by double exponential distributions; small jump sizes are modeled by a Normal distribution; jump intensities are constant
$\mathcal{M}_5$ (NEW)	LH jump sizes are modeled by double exponential distributions; small jump sizes are modeled by a Normal distribution; jump intensities are modeled by a mutually exciting process

Note: This table describes the specifications for the jump components in each model. Models $\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_3$ are classical model proposed by Heston (1993), Bates (1996) and Kou and Wang (2004), respectively. $\mathcal{M}_4$ and $\mathcal{M}_5$ are new models proposed in this study.

Table 3
DIC and LLK of different models.

Model	002327	002185	000720	All	002327	002185	000720	All
PLH(%)	0.59%	2.35%	6.20%	3.07%	0.59%	2.35%	6.20%	3.07%
	DIC				LLK			
NGARCH	-	-	-	-	0.80	0.97	0.81	0.85(0%)
$\mathcal{M}_1$	1.00	1.00	1.00	1(0%)	1.00	1.00	1.00	1(0%)
$\mathcal{M}_2$	0.94	0.98	0.96	0.99(%)	1.03	1.02	1.01	1.02(0.5%)
$\mathcal{M}_3$	0.96	0.99	0.93	0.99(9.2%)	1.00	1.01	1.02	1.01(11.5%)
$\mathcal{M}_4$	**0.89**	0.91	0.92	0.98(20.6%)	1.04	1.00	1.02	1.03(16.3%)
$\mathcal{M}_5$	0.91	**0.89**	**0.93**	0.97(69.3%)	**1.06**	**1.02**	**1.05**	1.08(71.6%)

Note: This table presents the Deviance Information Criterion (DIC) and log-likelihood (LLK) of three example stocks with ID: 002327, 002185, and 000720. The second row, PLH(%), presents the percentage of price-limit-hitting events of three stocks during the data period. The percentage is 3.07% for all stocks on average. Column "All" represents the average value of all stocks (and the percentage of stocks that have the lowest DIC and the highest LLK using the corresponding model). The values of DIC and LLK of $\mathcal{M}_1$ are set as 1, while those for other models are represented as ratios relative to $\mathcal{M}_1$.

PLH events are not that rare, with an average occurrence probability of 3.14% per year.[8]

3.2. Model performance

To evaluate the performance of our proposed model, we compare it (denoted by $\mathcal{M}_5$ in this subsection) with the Heston (1993) stochastic volatility (SV) model ($\mathcal{M}_1$), the SVJ model proposed in Bates (1996) (denoted by $\mathcal{M}_2$ in this subsection), the SV model with double exponential jumps by Kou and Wang (2004) ($\mathcal{M}_3$) and a model with mixed specifications on jumps ($\mathcal{M}_4$). $\mathcal{M}_1$ to $\mathcal{M}_4$ all assume constant jump intensity. Table 2 presents a summary of these five models. Notably, to our best knowledge, $\mathcal{M}_4$ and $\mathcal{M}_5$ are both new models proposed in this paper. In addition, we also consider a classic non-affine discrete-time model, the nonlinear asymmetric generalized autoregressive conditional heteroskedasticity (NGARCH) model (Engle and Ng, 1993), as a benchmark.

For a preliminary comparison of model fit, we select three stocks based on the number of price limit hits.[9] We plot stock returns, estimated volatility, and return minus estimated jumps of these three stocks for different models in Fig. 1. As shown in Fig. 1, $\mathcal{M}_1$ and $\mathcal{M}_2$ explain primarily the variations in returns through the variance component. However, this is less likely to be a valid way, particularly when there are numerous LH events. In addition, it is essential to model LH jumps and other small jumps separately.

In Table 3, we provide the log-likelihood (LLK) and deviance information criterion (DIC) values for the different models, which serve as measures of model fitness. The values of $\mathcal{M}_1$ are set as 1, while those for other models are represented as ratios relative to $\mathcal{M}_1$. A higher LLK value and a lower DIC value indicate a better fit for the data. It is worth noting that DIC measures the model fitness by penalizing the fit of a model by its complexity, determined by the effective number of parameters.

In addition to the three selected stocks, we also present the percentage of all stocks for which each model has the lowest DIC and highest

[8] This PLH frequency appears to be sufficient for a precise estimation. Other studies examining the S&P 500 index often report lower jump intensity estimates. For example, 0.66% in Eraker et al. (2003), 0.6% in Li et al. (2008), and 0.9% in Neumann et al. (2016), also see Kou and Wang (2004) and Kou et al. (2017) for comparable estimates.

[9] We sort all Chinese A-share stocks by the number of price limit hits and choose stocks at the 10th, 50th, and 90th percentiles. Three selected stocks' tickers are 002327, 002185, and 000720. Note that the third stock (000720) has two layers of price limits, with a 10% limit for approximately half of the time and a 5% limit for the other half when it was classified as an ST stock.

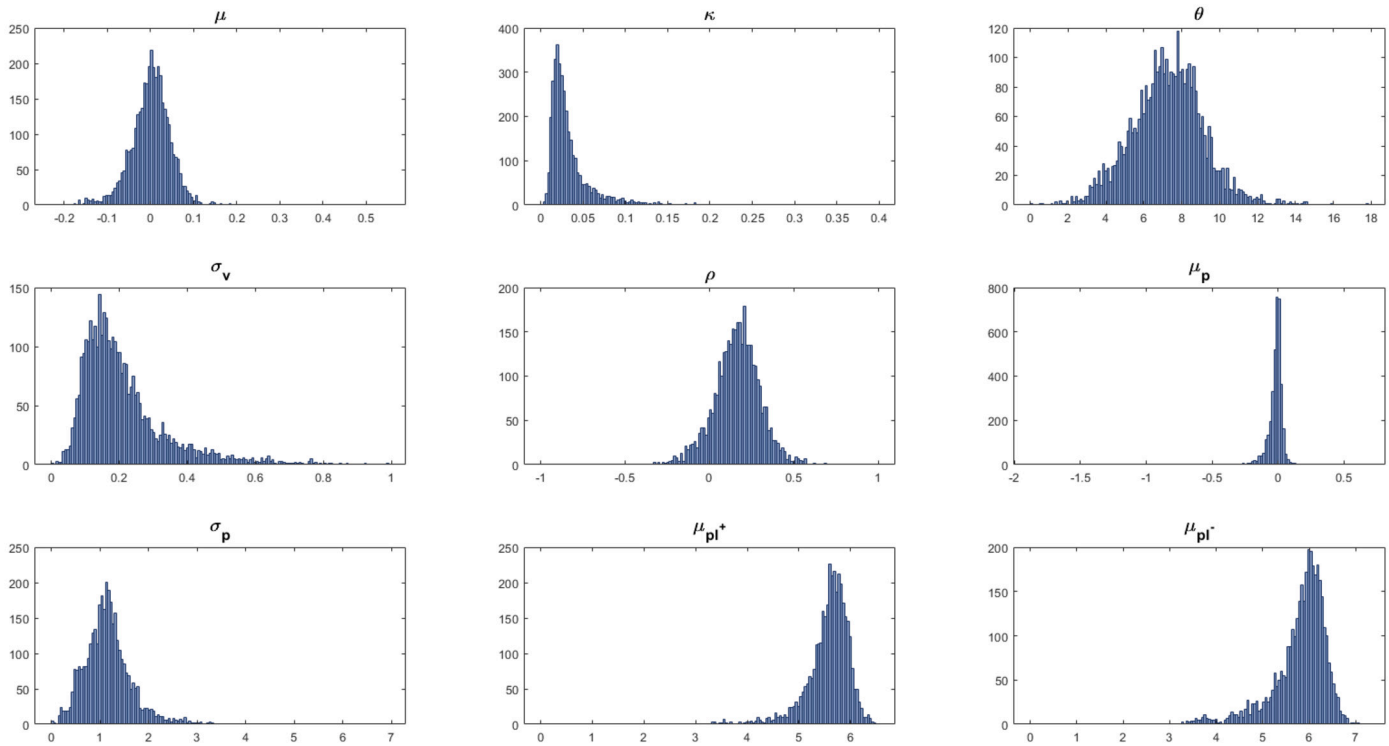

Fig. 2. Histograms of static parameters in the return and variance process. The x-axis represents the posterior means of the estimated parameters while the y-axis represents the number of companies.

LLK values.[10] The classical models, NGARCH, $\mathcal{M}_1$, $\mathcal{M}_2$, and $\mathcal{M}_3$, generally exhibit inferior performance compared to $\mathcal{M}_4$ and $\mathcal{M}_5$. It is worth noting that the basic affine model, SV, reveals much better fitness in the China stock market compared to the non-affine model, NGARCH. The two new models, $\mathcal{M}_4$ and $\mathcal{M}_5$, proposed in this study provide the best fit for more than 88% of the stocks in our data sample. This demonstrates the superiority of our proposed models. Furthermore, the results emphasize the importance of considering the interdependence among jumps, as $\mathcal{M}_5$ outperforms other models in fitting approximately 70% of the stocks.

To further investigate the model fitness of different models, we categorized all stocks into different portfolios based on the frequency of price limits hitting (PLH) events. The results are presented in Table 4. As is shown in Table 4, our model especially performs better when there are more PLH events in the data. This underscores the imperative to consider the LHE effect in a model when the PLH events are more frequent (a number of 2% implies about 5 PLH events per year). We also categorized stocks based on their respective industry classification, and $\mathcal{M}_5$ consistently outperforms other models across all industries. Details of these results can be found in Table A1 of the online supplementary appendix.

Table 4
DIC and LLK of PLH%-sorted portfolio.

portfolio	1	2	3	4	5
PLH(%)	1.23%	2.03%	2.68%	3.49%	6.29%
	DIC				
$\mathcal{M}_1$	0.0%	0.0%	0.0%	0.0%	0.0%
$\mathcal{M}_2$	0.0%	0.1%	0.0%	0.3%	0.4%
$\mathcal{M}_3$	4.8%	2.0%	1.3%	0.8%	0.1%
$\mathcal{M}_4$	9.9%	6.7%	3.5%	2.2%	0.2%
$\mathcal{M}_5$	5.2%	11.2%	15.1%	16.7%	19.3%
Sum	20%	20%	20%	20%	20%
	LLK				
NGARCH	0.0%	0.0%	0.0%	0.0%	0.0%
$\mathcal{M}_1$	0.0%	0.0%	0.0%	0.0%	0.0%
$\mathcal{M}_2$	0.0%	0.0%	0.0%	0.3%	0.5%
$\mathcal{M}_3$	2.0%	3.0%	3.4%	1.0%	0.1%
$\mathcal{M}_4$	12.1%	4.5%	1.0%	0.0%	0.0%
$\mathcal{M}_5$	6.0%	12.5%	15.5%	18.5%	19.4%
Sum	20%	20%	20%	20%	20%

Note: This table presents the Deviance Information Criterion (DIC) and log-likelihood (LLK) of the PLH%-sorted portfolios. We sort all stocks into five portfolios according to PLH% and report the percentage of stocks that have the lowest DIC and the highest LLK using the corresponding model. PLH(%) is the average percentage of PLH events across stocks in the portfolio.

3.3. Analysis of estimated parameters

This section offers some initial analysis of the estimated results. Fig. 2 presents the posterior means' histogram of stocks' static parameters in the return and variance processes, and Fig. 3 displays the histogram of the estimated $\boldsymbol{\lambda_0}$ and $\boldsymbol{\alpha}$ for all companies we used. Based on the estimates of $\boldsymbol{\lambda_0}$, one can find that LH events are relatively rare compared with small jumps. However, there are more stocks whose limit hits have a significant subsequent impact on future jumps than small jumps.

Based on the estimates of $\boldsymbol{\alpha}$, it is evident that limit-hitting jumps are more likely to trigger subsequent jumps. This is indicated by the higher estimates of incremental intensities (α) associated with LH jumps compared to small jumps. Moreover, both upper-limit-hitting and lower-limit-hitting jumps exhibit relatively strong self-exciting effects, suggesting that the occurrence of an upper (lower) limit hit increases the probability of another upper (lower) limit hit in the future. By contrast, the probability of inversely hitting lower (upper) limits after a(n) upper (lower) limit hit is much lower.

[10] Following Duan and Zhang (2014), the NGARCH model is estimated using the generalized method of moments (GMM) method. Thus, only the LLK of the NGARCH model is applied in the model comparison.

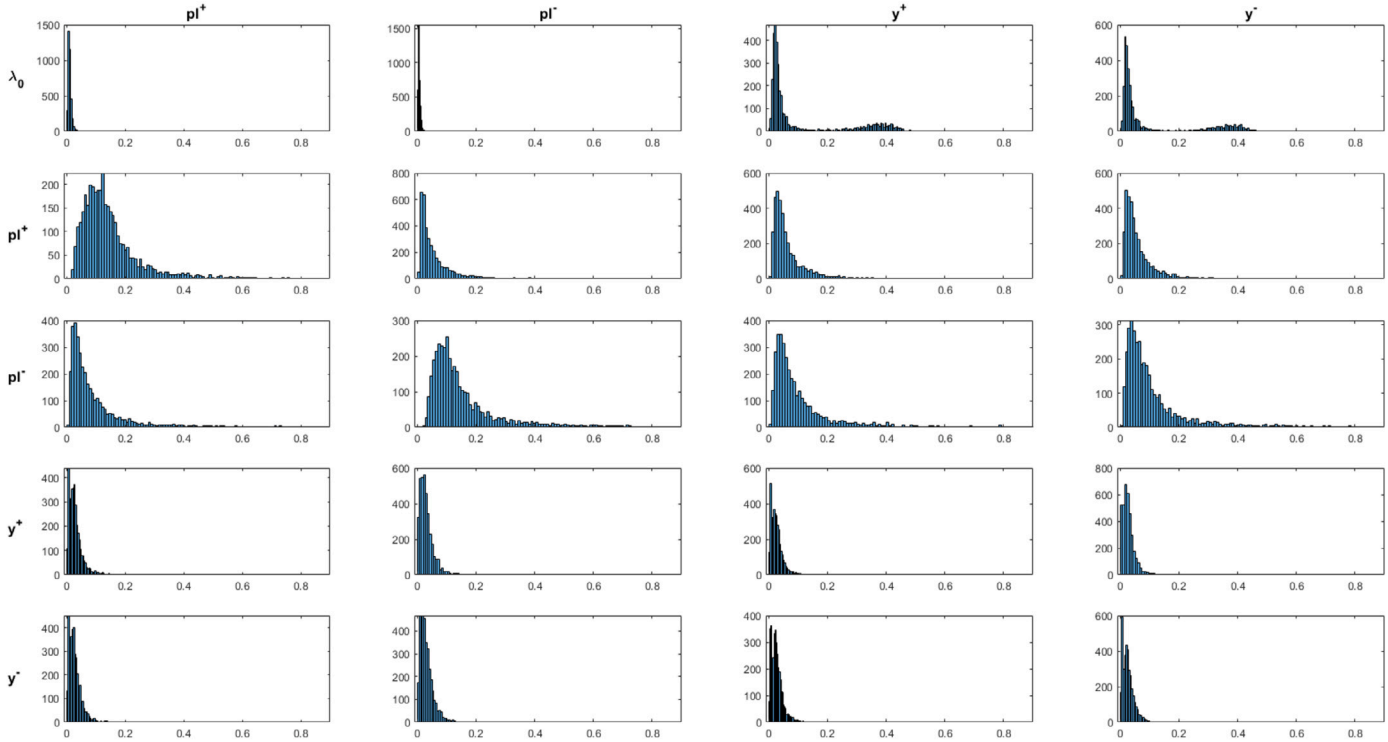

Fig. 3. Histograms of $\boldsymbol{\lambda_0}$ and $\boldsymbol{\alpha}$. The x-axis represents the posterior means of the estimated parameters while the y-axis represents the number of companies. The dimensions from the second to the fourth row correspond to j, and the dimensions from the first to the fourth column correspond to i. For example, the small graph in the second row and second column represents α_{pl^+,pl^-}, which measures the incremental intensity of pl^- after hitting the upper limit.

Table 5
Significance of estimated parameters.

	j \ i	pl^+	pl^-	y^+	y^-
λ_0^i		100.0%	100.0%	99.8%	99.8%
$\alpha_{j,i}$	pl^+	95.3%	67.3%	21.1%	19.4%
	pl^-	75.5%	92.8%	32.1%	35.0%
	y^+	4.0%	2.9%	2.6%	3.5%
	y^-	3.0%	4.5%	3.2%	2.8%
$\beta_{j,i}$	pl^+	96.9%	85.2%	99.5%	98.8%
	pl^-	86.7%	96.6%	99.0%	99.3%
	y^+	99.6%	99.8%	99.2%	98.5%
	y^-	99.8%	99.8%	98.3%	99.3%

Note: This table presents the percentage of stocks whose estimated posterior means of corresponding parameters minus two times their posterior standard deviations is greater than 0.

Furthermore, the results indicate that not all types of jumps exhibit noticeable jump-exciting effects in the Chinese stock market - only those jumps that drive the stock price to close at limits, which is a limit-hitting event, reveal strong jump-exciting effects. This is in stark contrast to those markets which have no price limit mechanism. Moreover, compared to upper limit hits, lower limit hits can raise a higher probability of future small jumps. The average values of estimated α^{pl^+,y^+} and α^{pl^+,y^-} are smaller than those of α^{pl^-,y^+} and α^{pl^-,y^-}. The estimates of α^{pl^-,y^+} and α^{pl^-,y^-} even reach 0.8 for some stocks, suggesting that a lower limit hit today increases the probability of small price jumps tomorrow by 0.8. Conversely, the estimates of α^{pl^+,y^+} and α^{pl^+,y^-} are all smaller than 0.4.

Table 5 presents information on the significance of the estimated MHP-related parameters. The majority of the estimated $\boldsymbol{\lambda_0}$ and $\boldsymbol{\beta}$ are significantly different from zero. However, the estimates of $\boldsymbol{\alpha}$, which quantify the incremental intensity results from jumps, are generally significant solely in the dimensions of pl^+ and pl^-.

Specifically, the results indicate that for less than 5% of companies, small jumps have a significant impact on increasing the probability of subsequent price jumps. However, for approximately 65% to 95% of companies, hitting price limits significantly increases the likelihood of subsequent limit hits. Moreover, for more than 19% of companies, hitting price limits significantly raises the probability of subsequent small jumps. These findings suggest that while there may be numerous small jumps, it is the jumps reaching the price limits that have a stronger impact and are more likely to trigger subsequent price jumps.

In summary, the estimation results highlight that LH jumps have a more pronounced and statistically significant impact in triggering price jumps than small jumps. Consistent with the three hypotheses regarding price limit critics discussed in Section 1, we quantitatively show that significant LHE effects exist in the financial market that adheres to daily price limit rules.

3.4. Impact of firm characteristics on jump clustering effects

To further investigate the impact of different firm characteristics on jump clustering effects, we conduct a correlation analysis using three firm-level indicators: firm sizes, turnover ratios and holding period returns; the results are reported in Table 6.

In Panel A of Table 6, we examine the correlations between the estimated parameters and firm sizes. Consistent with Seasholes and Wu (2007) we find that the likelihood of hitting price limits ($\lambda_0^{pl^+}$ and $\lambda_0^{pl^-}$) is negatively associated with firm sizes. This underpins the findings in Leippold et al. (2022) that large stocks tend to be more liquid with lower price volatility, and are therefore less affected by daily price limits in the Chinese market. However, the stock prices of large firms tend to bear more small jumps.

While hitting price limits is infrequent for large stocks, our results reveal that these firms exhibit stronger self and mutually exciting ef-

Table 6
Correlation analysis.

Panel A. Correlations between estimated parameters and firm sizes					
	j \ i	pl^+	pl^-	y^+	y^-
λ_0^i		-0.080***	-0.016	0.093***	0.092***
$\alpha_{j,i}$	pl^+	0.059***	0.084***	0.097***	0.110***
	pl^-	0.147***	0.063***	0.099***	0.093***
	y^+	-0.062***	-0.039**	-0.087***	-0.082***
	y^-	-0.064***	-0.051***	-0.094***	-0.082***
$\beta_{j,i}$	pl^+	0.104***	0.042***	-0.041***	-0.005
	pl^-	0.008	0.075	-0.027	-0.033**
	y^+	0.002	0.011	-0.017	-0.04**
	y^-	-0.002	-0.007	-0.04***	-0.017
Panel B. Correlations between estimated parameters and turnover ratio					
λ_0^i		0.354***	0.312***	0.150***	0.155***
$\alpha_{j,i}$	pl^+	0.148***	0.062***	0.002	0.025
	pl^-	0.115***	0.188***	0.158***	0.173***
	y^+	0.034**	0.105***	0.018	0.019
	y^-	0.02	0.077***	0.019	-0.006
$\beta_{j,i}$	pl^+	0.022	0.076***	0.02	0.016
	pl^-	0.210***	0.325***	-0.009	-0.014
	y^+	-0.012	0.019	-0.003	-0.013
	y^-	-0.019	-0.005	0.014	-0.018
Panel C. Correlations between estimated parameters and returns					
λ_0^i		0.036**	-0.057***	-0.128***	-0.150***
$\alpha_{j,i}$	pl^+	-0.212***	-0.175***	-0.181***	-0.231***
	pl^-	-0.158***	-0.172***	-0.159***	-0.151***
	y^+	0.011	-0.035**	0.024	0.033**
	y^-	0.023	0.005	0.056***	0.091***
$\beta_{j,i}$	pl^+	-0.121***	-0.055***	-0.012	0.057***
	pl^-	-0.106***	-0.115***	0.012	0.055***
	y^+	-0.005	-0.002	0.01	0.005
	y^-	0.006	0.021	0.02	0.028*

Note: This table presents correlations between estimated MHP-related parameters and different firm-level indicators. Panel A, B and C report the correlations between estimated MHP-related parameters and firm sizes, stock turnover ratio and stock daily average returns, respectively. *, ** and *** indicate values significant at 10%, 5% and 1% significance levels, respectively.

fects. This leads to a higher probability of subsequent price jumps (both small jumps and LH jumps) following LH events. This is reasonable as hitting price limits attracts more attention when such events are uncommon.

Baker and Wurgler (2006) emphasize that the turnover ratio can serve as a proxy to measure investors' sentiment. In the subsequent study, Baker and Wurgler (2007) suggest that investor sentiment contributes to financial market crashes. Based on the correlations reported in Panel B of Table 6, we find that the turnover ratio of companies is significantly and positively linked to the frequency of LH events. This result implies a potential link between investor sentiment and LH occurrences.

Additionally, we find that the jump-exciting effects for LH jumps are stronger for companies with a higher turnover ratio. This implies that LH events have the potential to stimulate investors' sentiment and produce more subsequent price jumps. By contrast, the jump-exciting effects for small jumps, which are not apparent in a market subject to price limits, do not consistently correlate significantly with the stock turnover ratio.

Panel C of Table 6 reports the correlations between the estimated MHP-related parameters and the returns for all stocks (firms' returns are their daily average returns during the study period). The results show that the intensity of hitting upper (lower) limits is significantly and positively (negatively) linked to firms' returns; however, the correlation sizes are both small (less than 0.1). Additionally, both the positive and negative small jump frequency is negatively linked to the firms' returns.

Notably, the estimates of incremental intensity (α) for LH jumps reveal the most compelling relationship with firms' mean returns. The firms' returns are significantly and negatively associated with the incremental jump probabilities following limit hits. Furthermore, the decaying speed of the incremental intensity for limit hits also exhibits a significant correlation with firms' returns. In comparison, the jump clustering effects of small jumps merely correlate significantly with stock returns.

4. Clustering index and preliminary analysis

In this section, we propose a clustering index (CI) that measures the jump clustering effects of firms and conduct further analysis based on the CI.

The clustering index is defined as follows:

$$CI_{j,i} = \int_0^{+\infty} \alpha_{j,i} e^{-\beta_{j,i} t} dt = \frac{\alpha_{j,i}}{\beta_{j,i}}. \tag{9}$$

The intuition behind this index lies in examining the expected number of jumps in dimension i resulting from jump occurrences in dimension j. It is convenient to assess the clustering effect caused by specific or combinations of dimensions using the clustering index directly.

To demonstrate the realized behaviors of limit hits for stocks with varying levels of LHE effects, here we define $LHE = \sum CI_{j,i},\ j \in \{pl^+, pl^-\},\ i \in \{pl^+, pl^-, y^+, y^-\}$. We sort all stocks into five portfolios according to LHE. In Fig. 4, we plot the empirical conditional probability of PLH events at time t given a PLH event occurred at time $t-i$, $p(PL_t = 1 | PL_{t-i} = 1)$, $i = 1, 2, ..., 10$, in the left diagram. For a specific example, $p(PL_t = 1 | PL_{t-1} = 1) = 10\%$ means that there is 10% chance to observe a PLH event at time t given that a PLH event occurred at time $t-1$. The right diagram presents the mean absolute returns (excluding returns that hit price limits) at time t given a PLH event at time $t-i$. These two figures show a clear clustering effect following PLH events, characterized by a heightened probability of another PLH and an increased magnitude of subsequent returns.

4.1. Clustering index

Table 7 presents an overview of the clustering index levels. In Panel A, we present the mean CI values for all stocks in our research sample. Small jumps occur more frequently than limit-hitting jumps, with the baseline intensity for small jumps being 13 times greater than that for limit hits. However, LH jumps are more likely to trigger subsequent price jumps, as indicated by the higher values reported in columns (4) to (7) compared to columns (8) to (11). This is because investors do not easily recognize small jumps, while LH jumps are more obvious. Prior studies (e.g., Seasholes and Wu, 2007; Chen et al., 2019 and Lin et al., 2023) have also found that stocks hitting price limits attract investors' attention.

We investigate two types of jump clustering effects: *amplified jump clustering* and *reversal jump clustering*. The amplified jump clustering effect occurs when a jump is followed by subsequent jumps in the same direction (e.g., $pl+$ to $pl+$ or $y+$), while the reversal jump clustering effect occurs when a jump is followed by subsequent jumps in the opposite direction (e.g., $pl+$ to $pl-$ or $y-$). Regarding LH jumps, the amplified jump clustering effect suggests that market information is not fully reflected because of the price limits mechanism, leading to a higher probability of subsequent jumps in the same direction. Conversely, the reversal jump clustering effect suggests market overreactions to information or the presence of possible market manipulations (Chen et al., 2019).

Our results indicate that LH jumps are more likely to trigger subsequent jumps in the same direction, implying that most limit hits occur due to the existence of price limit mechanisms, which hinder the full reflection of market information in time.

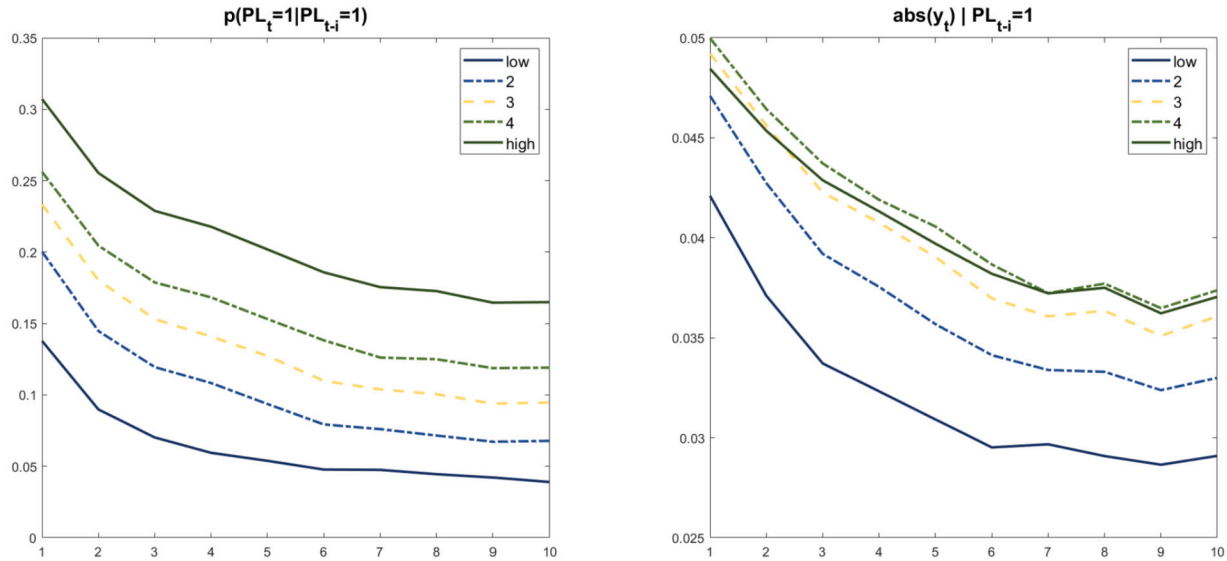

Fig. 4. Realized behaviors of limit hits. We sort all stocks into five portfolios according to the level of LHE effects. The left figure presents the empirical conditional probability of PLH events at time t given a PLH event occurred at time $t-i$, $i = 1, 2, ..., 10$. The right figure presents the mean absolute returns at time t given a PLH event occurred at time $t-i$. When calculating these returns, we exclude returns that hit price limits.

Table 7
Mean estimates of clustering index.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	#	λ_0^{pl}	λ_0^{y}	$CI_{pl^+,R}$	$CI_{pl^-,R}$	$CI_{pl^+,A}$	$CI_{pl^-,A}$	$CI_{y^+,R}$	$CI_{y^-,R}$	$CI_{y^+,A}$	$CI_{y^-,A}$
Panel A. Mean Estimates for All Samples											
All	3646	0.0175	0.2262	0.2155	0.1542	0.3479	0.4058	0.0979	0.0664	0.0600	0.0872
Panel B. Mean Estimates for Different Industries											
Finance	108	0.0177	0.2540	0.2225	0.1601	0.3646	0.3453	0.1525	0.1139	0.0807	0.1258
Public Utility	649	0.0187	0.2140	0.2354	0.1481	0.3600	0.4068	0.0980	0.0661	0.0585	0.0844
Properties	185	0.0150	0.1115	0.2304	0.1599	0.3716	0.4312	0.0773	0.0602	0.0487	0.1094
Conglomerates	83	0.0164	0.1407	0.2404	0.1797	0.3393	0.4293	0.0749	0.0492	0.0684	0.0678
Industrials	2443	0.0177	0.2454	0.2075	0.1520	0.3419	0.4026	0.0995	0.0666	0.0609	0.0862
Commerce	178	0.0141	0.1792	0.2218	0.1869	0.3554	0.4452	0.0747	0.0513	0.0495	0.0740
Panel C. Mean Estimates for SOE and PE											
SOE	1125	0.0150	0.1385	0.2409	0.1621	0.3363	0.4464	0.0802	0.0549	0.0506	0.0769
PE	2521	0.0187	0.2675	0.2042	0.1507	0.3531	0.3877	0.1059	0.0716	0.0642	0.0918

Note: This table presents the mean estimates of the clustering index. Panel A presents the mean values across all stocks. Panel B presents the corresponding mean estimates for different industries. The CSMAR classifies stocks into six industry groups: finance, public utility, properties, conglomerates, industrials, and commerce. Panel C shows the corresponding values for state-owned enterprises (SOE) and private enterprises (PE). The number of stocks is given in column (1); $\lambda_0^{pl} = \lambda_0^{pl^+} + \lambda_0^{pl^-}$ represents the base-line intensity for limit hits; $\lambda_0^{y} = \lambda_0^{y^+} + \lambda_0^{y^-}$ represents the base-line intensity for small jumps. We focus on two kinds of clustering effect: the amplified jump clustering effect (denoted by A) describes the situation that a jump will lead to subsequent jumps in the same direction, and the reversal jump clustering effect (denoted by R) describes the situation that a jump will lead to subsequent jumps in the opposite directions. Columns (4) to (11) show $CI_{pl^+,pl^-} + CI_{pl^+,y^-}$, $CI_{pl^-,pl^+} + CI_{pl^-,y^+}$, $CI_{pl^+,pl^+} + CI_{pl^+,y^+}$, $CI_{pl^-,pl^-} + CI_{pl^-,y^-}$, $CI_{y^+,pl^-} + CI_{y^+,y^-}$, $CI_{y^-,pl^+} + CI_{y^-,y^+}$, $CI_{y^+,pl^+} + CI_{y^+,y^+}$ and $CI_{y^-,pl^-} + CI_{y^-,y^-}$, respectively.

A heterogeneity analysis is also conducted for different industries, and the results are presented in Panel B of Table 7. The CI values for LH jumps are found to be quite similar across different industries. This suggests that the attention-grabbing nature of price limit hits has a consistent impact regardless of the industry. However, the Finance industry exhibits a significantly stronger clustering effect for small jumps, as shown in columns (8) to (11). This can be attributed to the higher level of attention in the Finance industry, making small price jumps more noticeable and leading to subsequent price jumps. Additional evidence supporting this observation is provided in Table A2 of the online supplementary appendix, which shows that companies in the finance industry have nearly twice as much analyst coverage and media coverage compared to companies in other industries. Furthermore, they have a higher proportion of institutional investors holding their stocks. These factors contribute to the increased attention and heightened sensitivity to small price movements in the finance industry, resulting in a stronger clustering effect for small jumps.

The CI values for state-owned enterprises (SOE) and private enterprises (PE) are listed in Panel C of Table 7. The results suggest that the SOE stock price is less prone to jumps, with the baseline intensity for small jumps being only half of that for PE stocks. Additionally, the results show that limit-hitting events, particularly lower limit hits, tend

Table 8
Portfolio analysis: amplified LH jumps clustering effects.

	CI	Mkt Cap (10^{-6})	Trd Amt (10^{-8})	AR ($*$ 252)	TO ratio	Std Dev ($*\sqrt{252}$)	Skew	Kurt	Amihud (10^{6})
min	0.05	9.12	1.69	21.6%	26.91	0.51	0.08	4.88	80.23
2	0.06	9.05	1.61	19.1%	28.77	0.51	0.09	5.00	98.79
3	0.07	11.42	1.59	18.0%	27.86	0.51	0.09	5.16	71.76
4	0.08	8.96	1.52	17.5%	29.47	0.51	0.11	5.19	50.55
5	0.09	9.94	1.50	17.2%	30.75	0.51	0.13	5.43	36.24
6	0.10	10.60	1.48	17.0%	31.95	0.50	0.14	5.52	25.51
7	0.12	9.51	1.36	15.5%	34.21	0.49	0.17	5.61	12.50
8	0.15	15.52	1.62	12.4%	37.19	0.49	0.20	5.79	16.36
9	0.20	21.20	1.62	11.0%	40.73	0.47	0.20	5.95	11.62
max	0.32	26.26	1.73	7.8%	37.75	0.44	0.25	6.27	10.67

Note: This table reports various indicators of the portfolios sorted based on the clustering index. Regarding the amplified LH jumps clustering effects, $j \to i$ can be $pl^+ \to pl^+$ or y^+ for upper limits hitting jumps and $pl^- \to pl^-$ or y^-) for lower limits hitting jumps, and CI represents the average value. Mkt Cap denotes the market capital; Trd Amt represents the trading amount; AR represents the average daily return; TO ratio represents the turnover ratio; Std Dev, Skew and Kurt represent the second, third and fourth moments of firms' daily returns; Amihud represents the illiquidity ratio, a lower Amihud value indicates higher liquidity of the stock.

Table 9
Portfolio analysis: reversal LH jumps clustering effects.

	CI	Mkt Cap (10^{-6})	Trd Amt (10^{-8})	AR ($*$ 252)	TO ratio	Std Dev ($*\sqrt{252}$)	Skew	Kurt	Amihud (10^{6})
min	0.02	5.87	1.13	17.6%	27.73	0.52	0.08	4.62	146.96
2	0.03	7.54	1.45	20.3%	29.62	0.52	0.08	4.95	98.69
3	0.04	7.66	1.45	18.1%	30.56	0.52	0.10	5.29	185.06
4	0.04	9.24	1.65	17.7%	30.34	0.51	0.11	5.30	64.10
5	0.05	9.67	1.51	16.9%	29.93	0.50	0.12	5.42	72.69
6	0.07	11.67	1.54	15.8%	31.65	0.50	0.15	5.56	26.17
7	0.08	14.08	1.88	16.3%	33.72	0.49	0.18	5.70	11.71
8	0.11	15.36	1.67	14.2%	38.25	0.48	0.18	5.74	12.06
9	0.16	19.66	1.77	12.0%	37.72	0.47	0.22	5.85	10.92
max	0.29	30.83	1.67	8.0%	36.06	0.44	0.24	6.36	10.49

Note: This table presents various indicators of the portfolios sorted based on the clustering index. Regarding reversal LH jumps clustering effects, $j \to i$ can be $pl^+ \to pl^-$ or y^- for upper limits hitting jumps and $pl^- \to pl^+$ or y^+) for lower limits hitting jumps, and CI represents the average value. Mkt Cap denotes the market capital; Trd Amt represents the trading amount; AR represents the average daily return; TO ratio represents the turnover ratio; Std Dev, Skew and Kurt are the second, third and fourth moments of firms' daily returns; Amihud represents the illiquidity ratio, a lower Amihud value indicates higher liquidity of the stock.

to trigger stronger reactions in SOE stocks, leading to higher CI values for lower limit-hitting jumps. Conversely, small jumps in PE stocks tend to have higher CI levels.

4.2. Portfolio analysis

We perform a portfolio analysis using the estimated CI values. We first look at the LH jumps and sort the 3646 stocks based on their CI into ten portfolios. Tables 8 and 9 report the results for both amplified and reversal LH jumps clustering effects. According to Tables 8 and 9, we find that both amplified and reversal LH jump clustering effects reveal apparent trends across all the indicators we consider, and the trends are consistent. One can find that large-size firms tend to have higher CI of LH jumps.

Additionally, higher CI of LH jumps are also characterized by greater turnover ratio, return skewness, kurtosis and liquidity. It is intriguing to note that the volatility of the returns decreases with the CI of the LH jumps, which implies that the application of price limits may contribute to market stabilization. Importantly, we also find that portfolios' average returns are negatively linked to the CI of LH jumps. This is in line with findings in Lin et al. (2023), which show that stocks that frequently hit limits can attract investor attention and have lower returns. In comparison, no apparent trend exists between portfolios' returns and CI levels of small jumps, as shown in Table A3 of the online supplementary appendix.

5. Further analysis with annual LHE effects estimates

The above results are obtained using the whole sample ranging from 2007 to 2021. To study the LHE effects in depth, we re-run the estimation to obtain the firm-level annual estimates for the LHE effects. We then further investigate the determinants of LHE effects and their ability to explain next-year return moments.

5.1. Determinants of limit-hitting exciting effects

The results so far suggest that the market has significant limit-hitting exciting effects; in other words, the stock price tends to continue to jump in the subsequent days after hitting price limits. The CI for limit-hitting jumps also correlates with some firm characteristics and stock return moments. We now explore further the potential determinants of the LHE effects in the cross section of stocks.

We examine the influence of attention-grabbing related factors, studied in the literature, on LHE effects, such as analyst coverage (Hirshleifer et al., 2013), media coverage (Hayo and Neuenkirch, 2015) and institutional holding ratio (Chen et al., 2020); Jeon et al. (2022) highlight that these three variables remain the crucial determinants of the jump sensitivity. Moreover, we also consider stock trading volume (Barber and Odean, 2008) and market capital (Clarke et al., 2020), which are related to investor attention. In our regression analysis, we include stock return moments (specifically mean return, volatility, skewness,

Table 10
Regression results: determinants of limit-hitting exciting effects.

	(1)	(2)	(3)	(4)
VARIABLES	$CI_{pl^+,R}$	$CI_{pl^-,R}$	$CI_{pl^+,A}$	$CI_{pl^-,A}$
AC	0.139	-0.0387	0.0939	0.202
MC	-1.119***	-1.154***	-0.119	0.779**
InsH	-2.281*	-6.454***	0.976	-2.489
Trd Amt	1.082***	1.433***	1.638***	0.177
Mkt Cap	0.222	0.178	-1.150**	-0.639
AR	-0.0819***	-0.119**	-0.0793**	-0.166***
Std Dev	30.29	-96.96***	-50.00**	-25.38
Skew	-0.799***	-0.966*	-1.030***	-0.381
Kurt	0.162*	0.0224	0.228**	0.113
λ_0^{pl}	-0.0500***	-0.00504	0.0557***	-0.0294***
λ_0^{y}	0.476***	0.842***	0.497***	0.500***
Constant	-15.07***	-8.541	1.381	34.64***
# of Obs	34,091	34,091	34,091	34,091
R^2	6.7%	6.2%	5.4%	2.9%

Note: This table presents the regression results of Equation (10). AC (MC) represents the natural logarithm of one plus the total number of analysts' forecasts (media reports); InsH represents the institutional holding ratio of a firm; Trd Amt (Mat Cap) represents the natural logarithm of daily average trading amount (market capital); AR, Std Dev, Skew and Kurt are the mean, standard deviation, skewness and kurtosis of daily returns; $\lambda_0^{pl} = \lambda_0^{pl^+} + \lambda_0^{pl^-}$ denotes the base-line intensity for limit hits; $\lambda_0^{y} = \lambda_0^{y^+} + \lambda_0^{y^-}$ denotes the base-line intensity for small jumps. We focus on two kinds of clustering effect: the amplified jump clustering effect (denoted by A) describes the situation that a jump will lead to subsequent jumps in the same direction, and the reversal jump clustering effect (denoted by R) describes the situation that a jump will lead to subsequent jumps in the opposite directions. Columns (1) to (4) present the results for $CI_{pl^+,pl^-} + CI_{pl^+,y^-}$, $CI_{pl^-,pl^+} + CI_{pl^-,y^+}$, $CI_{pl^+,pl^+} + CI_{pl^+,y^+}$ and $CI_{pl^-,pl^-} + CI_{pl^-,y^-}$, respectively. ***, ** and * indicate values significant at 1%, 5% and 10% significance levels, respectively.

and kurtosis), the estimated intensity of LH jumps and small jumps as additional explanatory variables.

To obtain more accurate results, we use a rolling window of three years to estimate the clustering index for each stock in each year from 2007 to 2021. For example, we estimate the CI values for 2007 using the stock returns from January 2005 to December 2007. We subsequently run the following regression to investigate the determinants of LHE effects.

$$\begin{aligned} CI_{f,t} =& \alpha + \beta_1 AC_{f,t} + \beta_2 MC_{f,t} + \beta_3 InsH_{f,t} + \beta_4 TrdAmt_{f,t} \\ &+ \beta_5 MktCap_{f,t} + \beta_6 AR_{f,t} + \beta_7 StdDev_{f,t} + \beta_8 Skew_{f,t} \\ &+ \beta_9 Kurt_{f,t} + \beta_{10}\lambda_{0,f,t}^{pl} + \beta_{11}\lambda_{0,f,t}^{y} + \epsilon_{f,t}. \end{aligned} \tag{10}$$

Where $CI_{f,t}$ denotes the clustering index of firm f in year t. We use four clustering indices for limit-hitting jumps. To enhance the readability of the regression results, the values of the dependent variable CI have been multiplied by 100. The explanatory variables are obtained from the CSMAR database. $AC_{f,t}$ ($MC_{f,t}$) is the natural logarithm of one plus the total number of analysts forecasts (media report) of firm f in year t; $InsH_{f,t}$ is the institutional holding ratio of firm f, averaged quarterly in year t; $TrdAmt_{f,t}$ ($MktCap_{f,t}$) is the natural logarithm of daily average trading amount (market capital) of firm f in year t; $AR_{f,t}$, $StdDev_{f,t}$, $Skew_{f,t}$ and $Kurt_{f,t}$ are the annualized mean, standard deviation, skewness and kurtosis of daily return in year t for firm f. Moreover, $\lambda_{0,f,t}^{pl} = \lambda_{0,f,t}^{pl+} + \lambda_{0,f,t}^{pl-}$; and $\lambda_{0,f,t}^{y} = \lambda_{0,f,t}^{y+} + \lambda_{0,f,t}^{y-}$. The error term is denoted as $\epsilon_{f,t}$. The descriptive statistics of these variables are provided in Table A4 of the online supplementary appendix.

The regression results are presented in Table 10. The results suggest that analyst coverage (AC) reveals no significant impact on LHE effects. This is because AC primarily reflects analysts' expectations for a firm's future earnings and pays less attention to short-term stock price fluctuations.

Conversely, media coverage (MC) significantly reduces the occurrence of reverse jump clustering. This is possibly due to the extensive media exposure making it challenging for people to form views on the corresponding company's stock that are opposite to previous market trends. Conversely, when stocks hit their lower limit, media exposure significantly increases people's behavior of chasing the downward trend. This implies that MC can amplify the negative sentiment and contribute to a further decline in stock prices.

Institutional investors rarely exhibit irrational behavior of chasing the upward trend or engaging in abrupt selling. The institutional holding ratio (InsH) does not significantly impact the amplified jump clustering effects. Moreover, companies with a higher proportion of institutional investors exhibit significantly smaller reversal jump clustering effects. Chen et al. (2019) find that only large investors would sell on the second day after a price limit-up. Institutional investors, in comparison, demonstrate more restraint and vigilance in their trading behaviors in response to LH events.

Companies with higher trading amounts exhibit higher LHE effects. In contrast, the impact of market capitalization on most LHE effects is insignificant. Our results indicate that the probability of investors chasing the upward trend after a stock hits the upper limit is significantly lower for companies with higher market capitalization. The stock return moments also reveal a significant impact on LHE effects, particularly for the stock mean return, which is significantly negatively linked to all four LHE effect proxies.

Companies with a higher frequency of price limit-ups and limit-downs have a lower probability of experiencing negative price movements after hitting the limits. This is because LH events are common for these companies, hence investors are less inclined to sell their stocks immediately after a limit hit. On the contrary, investors may seize the opportunity of a limit-up to capitalize on the potential for further price increases in these stocks. Furthermore, the strength of the LHE effect increases with the frequency of small jumps.

Despite the significant effects between these variables and LHE effects, the LHE effect remains a unique indicator. The maximum R-squared value is only 6.7%, indicating that the selected variables only account for a limited portion of the variability in LHE effects.

5.2. Explaining future return moments

We also investigate the impact of LHE effects on future stock return moments using the following regression model:

$$Rm_{f,t+1} = \alpha + \boldsymbol{\beta}_{CI}\mathbf{CI_{f,t}} + \boldsymbol{\beta}_{Control}\mathbf{Control_{f,t}} + \epsilon_{f,t+1}, \tag{11}$$

where CI contains the clustering index of the amplified and reversal clustering effects of LH jumps studied in Section 5.1, and we include all the explanatory variables from regression (10) as control variables. Rm denotes return moments, including the mean return, standard deviation, skewness and kurtosis.

The regression results are reported in Table 11. Consistent with Lin et al. (2023) and the previous results in this study, we observe that LHE effects exhibit a significant negative relationship with future stock returns. Moreover, the results indicate that LHE effects have a significant impact on future stock return volatility, skewness, and kurtosis. Overall, these findings suggest that LHE effects contribute to shaping future stock return moments and highlight their potential usefulness in asset pricing.[11]

[11] The size of R^2 reaches 22.6% in explaining stocks' next year returns; This compares favorably with other studies, such as Ferreira and Santa-Clara (2011), which finds that the R^2 statistic reaches 22.44% when forecasting next year returns using the smooth earnings price. Additionally, Kadilli (2015) achieves a R^2 of 46.8% in predicting next-month returns across countries, utilizing the

Table 11
Regression results: explaining future return moments.

	(1)	(2)	(3)	(4)
VARIABLES	AR	Std Dev	Skew	Kurt
L.$CI_{pl^+,R}$	-0.266**	-0.0433**	-0.0520***	0.123***
L.$CI_{pl^-,R}$	-0.355***	-0.0141	0.00385	0.00713
L.$CI_{pl^+,A}$	-0.179**	-0.0923***	-0.0308***	0.157***
L.$CI_{pl^-,A}$	-0.0848	-0.0489***	-0.00867	0.0721**
L.AC	-0.258***	-0.105***	-0.0194***	-0.0598***
L.MC	1.687***	0.165***	0.0676***	-0.614***
L.InsH	7.147***	-0.194***	0.0913***	1.515***
L.Trd Amt	-0.334***	-0.118***	-0.0917***	0.217***
L.Mkt Cap	-1.966***	-0.0394***	0.0505***	0.311***
L.AR	-0.289***	0.0450***	-0.00450***	-0.0412***
L.Std Dev	0.426***	0.138***	-0.073***	-0.115***
L.Skew	1.123***	0.250***	-0.0290***	-0.475***
L.Kurt	0.0862***	0.0130***	0.00702***	0.0301***
L.λ_0^{pl}	0.416***	-0.0101	-0.0758***	0.0845
L.λ_0^{y}	3.562***	1.092***	-0.115***	-1.606***
Constant	30.87***	5.183***	1.363***	-3.072***
# of obs	29,985	29,985	29,985	29,985
R^2	22.6%	20.7%	9.4%	18.2%

Note: This table presents the regression results of Equation (11). AC (MC) represents the natural logarithm of one plus the total number of analysts' forecasts (media reports); InsH represents the institutional holding ratio of a firm; Trd Amt (Mat Cap) represents the natural logarithm of daily average trading amount (market capital); AR, Std Dev, Skew and Kurt are the mean, standard deviation, skewness and kurtosis of daily returns; $\lambda_0^{pl} = \lambda_0^{pl^+} + \lambda_0^{pl^-}$ represents the base-line intensity for limit hits; $\lambda_0^{y} = \lambda_0^{y^+} + \lambda_0^{y^-}$ represents the base-line intensity for small jumps. We focus on two kinds of clustering effect: the amplified jump clustering effect (denoted by A) describes the situation that a jump will lead to subsequent jumps in the same direction, and the reversal jump clustering effect (denoted by R) describes the situation that a jump will lead to subsequent jumps in the opposite directions. Columns (1) to (4) present the results for $CI_{pl^+,pl^-} + CI_{pl^+,y^-}$, $CI_{pl^-,pl^+} + CI_{pl^-,y^+}$, $CI_{pl^+,pl^+} + CI_{pl^+,y^+}$ and $CI_{pl^-,pl^-} + CI_{pl^-,y^-}$, respectively. ***, ** and * indicate values significant at 1%, 5% and 10% significance levels, respectively.

6. Further analysis with monthly LHE effects estimates

The obtained results thus far indicate that LHE effects reveal a significant relationship with future return moments, suggesting the potential utilization of the CI of LH jumps as a cross-sectional stock pricing factor. This section provides an out-of-sample portfolio analysis using LHE effects on a monthly basis. In addition, we investigate the relationship between LHE effects and investor sentiment.

6.1. Out-of-sample portfolio analysis using monthly results

We perform a simple out-of-sample portfolio analysis using LHE effects on a monthly basis. Due to computational limitations, estimating the CI using our proposed model on a monthly basis is computationally intensive. The results in Table 5 provide compelling evidence that LH jumps, compared to small price jumps, serve as more significant indicators of subsequent jumps, particularly for subsequent LH jumps. Therefore, we focus solely on LH jumps and estimate MHP-related parameters using a rolling window of three years on a monthly basis.

In the out-of-sample portfolio analysis, we sort stocks into deciles based on the values of an aggressive LHE effect. The results, as shown in Table 12, demonstrate that the negative relationship between LH jumps CI and asset return persists in the out-of-sample analysis. The portfolio exhibiting the highest CI demonstrates a significantly negative difference in returns compared to the portfolio with the lowest CI, amounting to 89 basis points. This suggests that the CI has the potential to serve as a pricing factor and contributes to return prediction. However, we will refrain from further exploring this viewpoint because it is not the main focus of this study. More importantly, our results show that despite the presence of heavily-clustered LH jumps and return jumps, which are often associated with negative or lottery-like characteristics, these stocks are less likely to be classified as lottery-like stocks. This is because lottery stocks typically possess small market capitalization and low stock prices (see e.g., Bali et al., 2011).

Table 12
Portfolio analysis: out-of-sample results.

	CI	EW Return (%)	VW Return (%)	Mkt Cap	Price
min	0.108	1.980	1.372	7.8	16.8
2	0.139	1.928	1.215	7.5	15.7
3	0.160	1.883	1.146	7.7	15.1
4	0.179	1.731	1.153	8.8	15.7
5	0.198	1.613	0.881	9.6	15.3
6	0.220	1.568	0.870	10.2	14.8
7	0.249	1.590	1.312	11.9	14.8
8	0.288	1.616	1.099	15.0	15.1
9	0.350	1.470	0.978	22.6	15.4
max	0.530	1.123	0.478	37.2	17.7
10-1		-0.857	-0.894		
		(-2.67)	(-1.73)		

Note: This table presents the out-of-sample results obtained by sorting portfolios based on the clustering index. $j \in \{pl^+, pl^-\}$ and $i \in \{pl^+, pl^-\}$. CI is the average value. VW Returns denotes value-weighted returns; EW Returns denotes equal-weighted returns; Mkt Cap denotes the market capital; Price is the average price of stocks in the portfolio. The number in the parentheses is the t-statistics.

6.2. Limit-hitting exciting effects and investor sentiment

Taking the stock turnover ratio as a proxy for sentiment, Section 3.4 examines the relationship between the LHE effects and the stock turnover ratio. However, the turnover ratio has been dropped from the Baker and Wurgler sentiment index due to the explosion of high-frequency trading in recent years.[12] To incorporate other dimensions of the sentiment measure, we consider the investor sentiment index in China, known as CICSI, proposed in Yi and Mao (2009). The CICSI is computed based on the methodology in Baker and Wurgler (2006) and also takes the characteristics of the Chinese market into consideration. The CICSI is constructed using many market-level factors, such as the closed-end fund discount, the number of IPOs, the consumer confidence index and the number of new investor accounts opened, etc.[13] As the sentiment index is for the whole market, so we also construct market-level LHE effects to perform the following analysis.

The monthly market-level LHE effects (amplified or reversal) are computed as the equal-weighted values across all stocks. For comparison, we also take limit hits as dependent variables; the market-level upper limit hits (ULH) and lower limit hits (LLH) proxies are constructed as the equal-weighted number of ULH and LLH across all stocks during the month.[14] We also consider some attention-related variables used in Chen et al. (2022) as control variables: analyst coverage (AC), media coverage (MC), abnormal trading volume (ATV) and extreme returns (ER).[15] Furthermore, the China economic policy uncertainty (EPU) in-

same regression model as ours. We also refer to Liu and Thomas (2000), Duan and Zhang (2014), Li et al. (2014) and Jiang et al. (2019) for further insights and comparative results.

[12] We thank the reviewers for pointing this out and suggesting the further analysis in this subsection.

[13] The CICSI data comes from the CSMAR database and is updated monthly.

[14] The value-weighted values are also tested, and the conclusions remain. The results are available on request.

[15] The market-level proxies are constructed based on the methods in Chen et al. (2022). Analyst coverage (AC): We first count the total number of analyst

Table 13
Regression results: LHE effects and sentiment.

	LHE_A	LHE_R	ULH	LLH
Constant	0.02	-0.13	-0.27	-0.18
Sent	-0.04	0.30***	0.61***	0.40***
AC	0.04	-0.08	-0.11*	-0.15*
MC	0.07	-0.34***	0.08	0.20**
ATV	0.15*	0.19**	0.46***	0.09
ER	-0.15*	0	0.04	0.01
EPU	0.47***	-0.29***	-0.14*	-0.12
# of obs	144	144	144	144
adj. R^2	20.4%	18.7%	44.4%	9.3%

Note: LHE_A and LHE_R are the market-level amplified and reversal LHE effects, respectively; ULH (LLH) are the natural logarithm of one plus the corresponding market-level values; AC (MC) represents the natural logarithm of one plus the corresponding market-level values; and ATV is the abnormal trading volume; ER represents the extreme return. Sent denotes the CICSI China market sentiment index, and EPU indicates the economic policy uncertainty. All variables are on a monthly basis. To facilitate comparisons, we standardize all variables. ***, ** and * indicate values significant at 1%, 5% and 10% significance levels, respectively.

dex is also considered as a control variable.[16] The following regression model is performed to study the linkage between investor sentiment and LHE effects:

$$y_t = \alpha + \beta_{Sent}\text{Sent}_t + \boldsymbol{\beta_{Control}Control_t} + \epsilon_t. \tag{12}$$

The results are reported in Table 13. Investor sentiment is significantly positively related to the reversal LHE effects. The reversal of LHE effects reflects investors' disagreement with stock prices, and the degree of disagreement increases with investor sentiment. In contrast, the amplified LHE effect is not significantly related to investor sentiment. In addition, investor sentiment is positively related to both ULH and LLH, suggesting that limit-hitting events are more frequent when investor sentiment is high. It should be clarified that the number of LH events and the level of LHE effects are different; the LHE effects measure the probability of a subsequent price jump due to an LH event. When investor sentiment rises due to some events, stocks in the market tend to hit limits as a reaction, but this does not mean that these LH events cause a continuation of the subsequent price trend. This simply confirms our findings that the reversal LHE effect is relatively stronger when investor sentiment is high, while there is no significant effect on the amplified LHE effect. Furthermore, ATV is the only factor that is significantly and positively related to both the amplified and the reversal LHE effects. EPU also shows a significant effect on LHE effects. The amplified effects increase with China's economic policy uncertainty, suggesting that investors tend to chase the rise and kill the fall when uncertainty is high, while the reversal effects decrease with uncertainty, suggesting that when economic policy uncertainty is high, investors are afraid to make trades that go against the original price trend. In addition, consistent with the results in Section 5.1, AC reveals no significant relationship with LHE effects while MC is only negatively significantly linked to the reversal LHE effects.

7. Conclusions and discussions

We propose a new stochastic volatility model specifically designed for stock markets that implement price limit mechanisms. Our model regards LH events as jumps and characterizes the magnitudes of LH jumps using a double-exponential distribution, while the sizes of other small price jumps are modeled using a normal distribution. We further employ a multivariate Hawkes process and capture the interdependence among different types of jumps. Through empirical analysis, our proposed model has shown superior fitness compared to other traditional models.

The adoption of our model has facilitated a quantitative study of LHE effects utilizing empirical analysis of 3646 A-share stocks in the Chinese market. Our findings indicate that the LH jumps can significantly increase the probability of the occurrence of price jumps in subsequent days, demonstrating the existence of significant LHE effects. Additionally, the correlation analysis reveals a negative association between LHE effects and firm sizes and returns, while showing a positive association with the turnover ratio.

We further construct a clustering index based on the MHP parameters and find that the CI values of LH jumps are significantly higher than those of small jumps. Moreover, the amplified LH jump clustering effects are stronger than the reversal LH jump clustering effects. Furthermore, our analysis reveals that finance companies exhibit the highest small jump clustering effects than companies in other industries. In addition, we examine the attention-related factors as determinants of LHE effects. Selected factors partially explain these LHE effects. Specifically, while analyst coverage lacks significant explanatory power, media coverage and institutional holding ratio are significantly and negatively associated with reversal LHE effects.

Our research offers important implications for risk management in markets posing price limits. We have demonstrated that LH events have a significant impact on subsequent price jumps, emphasizing the need to consider these effects when evaluating tail risk. Moreover, our proposed model can be effectively applied in the iterative techniques for multiperiod tail risk forecasting of stocks in markets with price limits. Furthermore, the versatility of our model opens up possibilities for its potential application in option pricing within markets that implement price limit mechanisms, which presents an avenue for future study.

CRediT authorship contribution statement

Jian Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shuyuan Qi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Data availability

Data will be made available on request.

Appendix A. Bayesian MCMC algorithm

We construct the following Bayesian MCMC framework to approximate the posterior distribution in Equation (8):

1. Sample static parameters at the ιth iteration:
   Draw $\Theta_1^{(\iota)}$ from $p\left(\Theta_1^{(\iota)}|p_t, \Theta_2^{(\iota-1)}, \Theta_3^{(\iota-1)}, ..., \Theta_k^{(\iota-1)}, \Upsilon_t^{(\iota-1)}\right)$,
   ⋮

forecasts for each stock within a month. Then, the equal-weighted number of analyst forecasts across all stocks is computed as the measure of the aggregate stock market. Media coverage (MC): We first count the total number of media reports for each stock during the month. Then, the equal-weighted number of media reports across all stocks is taken as the measure of the aggregate stock market. Abnormal trading volume (ATV): We first calculate the ratio of trading volume at the end of each month to the average over the previous one year for each stock. Then, the equal-weighted abnormal trading volume across all stocks is used as the market-level measure. Extreme returns (ER): We first calculate the ratio of returns at the end of each month to the average over the previous one year for each stock. Then, the equal-weighted extreme returns across all stocks are taken as the market-level measure.

[16] Downloaded at https://www.policyuncertainty.com/china_monthly.html.

Draw $\Theta_k^{(\iota)}$ from $p\left(\Theta_k^{(\iota)}|p_t,\Theta_1^{(\iota-1)},\Theta_2^{(\iota-1)},...,\Theta_{k-1}^{(\iota-1)},\Upsilon_t^{(\iota-1)}\right)$.

2. Sample jumps at the ιth iteration:
   for t = 1,2,..., T:
   Draw $J_t^{y+,(\iota)}$ from $p\left(J_t^{y+,(\iota)}|p_t,\Theta^{(\iota)},J_t^{y-,(\iota-1)},\Xi_t^{(\iota-1)},\upsilon^{(\iota-1)}\right)$,
   Draw $J_t^{y-,(\iota)}$ from $p\left(J_t^{y+,(\iota)}|p_t,\Theta^{(\iota)},J_t^{y+,(\iota)},\Xi_t^{(\iota-1)},\upsilon^{(\iota-1)}\right)$,
3. Sample sizes of jumps at the ιth iteration:
   for t = 1,2,..., T:
   Draw $\xi_t^{y+,(\iota)}$ from
   $p\left(\xi_t^{y+,(\iota)}|p_t,\Theta^{(\iota)},\xi_t^{y-,(\iota-1)},\xi_t^{pl+,(\iota-1)},\xi_t^{pl-,(\iota-1)},\mathbf{J}_t^{(\iota)},\upsilon^{(\iota-1)}\right)$,
   Draw $\xi_t^{y-,(\iota)}$ from
   $p\left(\xi_t^{y+,(\iota)}|p_t,\Theta^{(\iota)},\xi_t^{y+,(\iota)},\xi_t^{pl+,(\iota-1)},\xi_t^{pl-,(\iota-1)},\mathbf{J}_t^{(\iota)},\upsilon^{(\iota-1)}\right)$,
   Draw $\xi_t^{pl+,(\iota)}$ from
   $p\left(\xi_t^{y+,(\iota)}|p_t,\Theta^{(\iota)},\xi_t^{y+,(\iota)},\xi_t^{y-,(\iota)},\xi_t^{pl-,(\iota-1)},\mathbf{J}_t^{(\iota)},\upsilon^{(\iota-1)}\right)$,
   Draw $\xi_t^{pl-,(\iota)}$ from
   $p\left(\xi_t^{y+,(\iota)}|p_t,\Theta^{(\iota)},\xi_t^{y+,(\iota)},\xi_t^{y-,(\iota)},\xi_t^{pl+,(\iota)},\mathbf{J}_t^{(\iota)},\upsilon^{(\iota-1)}\right)$,
4. Sample variance at the ιth iteration:
   for t = 1,2,..., T:
   Draw υ_t from $p\left(\upsilon_t^{(\iota)}|p_t,\Theta^{(\iota)},\mathbf{J}_t^{(\iota)},\Xi_t^{(\iota)}\right)$

In the estimation, we set the total number of iterations as $N = 80,000$ and the first 30,000 iterations are regarded as a burn-in period to be discarded. Our prior settings follow Lazar and Qi (2022) and Chen et al. (2023). Section B and C provides more details about the estimation methods.

Appendix B. Priors for model parameters

The priors of parameters are consistent with the literature (Eraker et al., 2003) and are summarized here: $\mu \sim \mathbb{N}(0,100)$, $\kappa \sim \mathbb{N}(0,1)\mathbf{1}_{\kappa>0}$, $\theta \sim \mathbb{N}(0,1)\mathbf{1}_{\theta>0}$, $\rho \sim \mathbb{U}(-1,1)$, $\sigma_v^2 \sim \mathbb{IG}(2.5,0.1)$, $\mu_p \sim \mathbb{N}(0,100)$, $\sigma_p^2 \sim \mathbb{IG}(10,40)$, $\mu_{pl+(-)} \sim \mathbb{IG}(10,20)$, $\lambda \sim \mathbb{B}(2,40)$, $\lambda_0^i \sim \mathbb{N}(0,0.01)\mathbf{1}_{\lambda_0^i>0}$, $\alpha_{j,i} \sim \mathbb{N}(0,0.3)\mathbf{1}_{\alpha_{j,i}>0}$ and $\beta_{j,i} \sim \mathbb{N}(0,1)\mathbf{1}_{\beta_{j,i}>0}$. $\mathbb{N}$ denotes the normal distribution; $\mathbb{IG}$ refers to the Inverse Gamma distribution, $\mathbb{U}$ represents the standard Uniform distribution, and $\mathbb{B}$ stands for the Beta distribution.

Appendix C. MCMC estimation methods

Posterior distributions and updating algorithms for parameters and latent variables are detailed in this section.

General parameters:

Posterior for μ: $\mu|p,\upsilon,J,\xi,\Theta\backslash\{\mu\} \sim \mathbb{N}(\mathcal{A},\mathcal{B})$, where $\mathcal{B} = \mathcal{A}\frac{\Delta}{(1-\rho^2)}\sum_{t=0}^{T-1}\frac{H_{t+1}-\frac{\rho}{\sigma_v}D_{t+1}}{\upsilon_t} + \frac{c^*}{C^*}$, $\mathcal{A} = \left[\frac{\Delta^2}{(1-\rho^2)}\sum_{t=0}^{T-1}\frac{1}{\upsilon_t}+\frac{1}{C^*}\right]^{-1}$, $D_{t+1} = \upsilon_{t+1} - \upsilon_t - \kappa(\theta-\upsilon_t)\Delta$, and $H_{t+1} = p_{t+1} - p_t - J_{t+1}\xi_{t+1}$. $c^* = 0$ and $C^* = 100$ are hyperparameters of the prior of μ. Δ denotes the time interval.

Posterior for κ: $\kappa|p,\upsilon,J,\xi,\Theta\backslash\{\kappa\} \sim \mathbb{N}\left(\frac{\mathcal{A}}{\mathcal{B}},\sqrt{\frac{1}{\mathcal{B}}}\right)\mathbf{1}_{\kappa>0}$, where $\mathcal{B} = \frac{\Delta}{(1-\rho^2)\sigma_v^2}\sum_{t=0}^{T-1}\frac{(\theta-\upsilon_t)^2}{\upsilon_t} + 1$, $\mathcal{A} = \frac{1}{\sigma_v(1-\rho^2)}\sum_{t=0}^{T-1}\frac{(\theta-\upsilon_t)\frac{\upsilon_{t+1}-\upsilon_t}{\sigma_v}-\rho H_{t+1}}{\upsilon_t}$, and $H_{t+1} = p_{t+1} - p_t - \mu\Delta - J_{t+1}\xi_{t+1}$.

Posterior for θ: $\theta|p,\upsilon,J,\xi,\Theta\backslash\{\theta\} \sim \mathbb{N}\left(\frac{\mathcal{A}}{\mathcal{B}},\sqrt{\frac{1}{\mathcal{B}}}\right)\mathbf{1}_{\theta>0}$, where $\mathcal{B} = \frac{\kappa^2\Delta}{(1-\rho^2)\sigma_v^2}\sum_{t=0}^{T-1}\frac{1}{\upsilon_t}+1$, $\mathcal{A} = \frac{\kappa}{\sigma_v(1-\rho^2)}\sum_{t=0}^{T-1}\frac{\frac{D_{t+1}}{\sigma_v}-\rho H_{t+1}}{\upsilon_t}$, $D_{t+1} = \upsilon_{t+1} + (\kappa\Delta - 1)\upsilon_t$, and $H_{t+1} = p_{t+1} - p_t - \mu\Delta - J_{t+1}\xi_{t+1}$.

Posterior for $\sigma_v^2 \propto (\sigma_v^2)^{\frac{T+c^*+2}{2}} \times \pi(\sigma_v) \times \exp\left\{-\frac{1}{2}\frac{C^*}{\sigma_v^2}\right\}$, where $c^* = 2.5$, and $C^* = 0.1$ are hyperparameters of the prior of σ_v^2, and $\pi(\sigma_v) := \exp\left\{-\frac{1}{2}\sum_{t=0}^{T-1}\frac{\left(D_{t+1}-\rho\sigma_v(H_{t+1})\right)^2}{(1-\rho^2)\sigma_v^2\upsilon_t\Delta}\right\}$. $D_{t+1} = \upsilon_{t+1} - \upsilon_t - \kappa(\theta-\upsilon_t)\Delta$ and $H_{t+1} = p_{t+1} - p_t - \mu\Delta - J_{t+1}\xi_{t+1}$. The updating algorithm is: (1) for a given previous draw $\sigma_v^{(g)}$, draw $\sigma_v^{(g+1)}$ from the proposal density function $\mathbb{IG}(\mathcal{A},\mathcal{B})$, where $\mathcal{A} = c^* + T$ and $\mathcal{B} = C^* + \sum_{t=0}^{T-1}\frac{D_{t+1}^2}{\upsilon_t\Delta}$, (2) accept $\sigma_v^{(g+1)}$ with probability $\min\left(\frac{\pi(\sigma_v^{(g+1)})}{\pi(\sigma_v^{(g)})},1\right)$.

Posterior for $\rho \propto \pi(\rho) := (1-\rho^2)^{-\frac{T}{2}}\exp\{-12(1-\rho^2)\sum_{t=0}^{T-1}(H_{t+1}^2 + D_{t+1}^2) + \frac{\rho}{1-\rho^2}\sum_{t=0}^{T-1}H_{t+1}D_{t+1}\}$, where $D_{t+1} = \frac{\upsilon_{t+1}-\upsilon_t-\kappa(\theta-\upsilon_t)\Delta}{\sigma_v\sqrt{\upsilon_t\Delta}}$ and $H_{t+1} = \frac{p_{t+1}-p_t-\mu\Delta-J_{t+1}\xi_{t+1}}{\sqrt{\upsilon_t\Delta}}$. The estimation is updated using the following algorithm: (1) Draw $\frac{1}{2}\ln\frac{1+\rho^{(g+1)}}{1-\rho^{(g+1)}}$ from $\mathbb{N}\left(\frac{1}{2}\ln\frac{1+\rho_r}{1-\rho_r},\frac{1}{T-3}\right)$, where $\rho_r = Corr(\mathbf{D},\mathbf{H})$, $\mathbf{D} = \{D_{t+1}\}_{t=0}^{T-1}$, $\mathbf{H} = \{H_{t+1}\}_{t=0}^{T-1}$, and $Corr$ denotes correlation; (2) accept ρ^{g+1} with $\min\left(\frac{\pi(\rho^{(g+1)})}{\pi(\rho^{(g)})} \times \frac{\exp(-\frac{(f(\rho^{(g)})-f(\rho_r))^2}{\frac{2}{T-3}})}{\exp(-\frac{(f(\rho^{(g+1)})-f(\rho_r))^2}{\frac{2}{T-3}})},1\right)$, where $f(\rho) = \frac{1}{2}\ln\frac{1+\rho}{1-\rho}$.

Posterior for J: $J_{t+1}|p,\upsilon,\xi^y,\Theta \sim Bernoulli\left(\frac{\alpha_1}{\alpha_1+\alpha_2}\right)$. $Bernoulli$ denotes the Bernoulli distribution. $\alpha_1 = \lambda_t\exp\left(-\frac{1}{2(1-\rho^2)}(H_{t+1}^2 - 2\rho H_{t+1}D_{t+1})\right)$, $D_{t+1} = \frac{\upsilon_{t+1}-\upsilon_t-\kappa(\theta-\upsilon_t)\Delta}{\sigma_v\sqrt{\upsilon_t\Delta}}$, $H_{t+1} = \frac{p_{t+1}-p_t-\mu\Delta-\xi_{t+1}^y}{\sqrt{\upsilon_t\Delta}}$, $\alpha_2 = (1-\lambda_t)\exp\left(-\frac{1}{2(1-\rho^2)}\left(HH_{t+1}^2 - 2\rho HH_{t+1}D_{t+1}\right)\right)$ and $HH_{t+1} = \frac{p_{t+1}-p_t-\mu\Delta}{\sqrt{\upsilon_t\Delta}}$. It is worth noting that the J values for different dimensions (pl^+,pl^-,y^+,y^-) can be easily distinguished based on the corresponding return direction and whether the price limits are reached.

Posterior for $\upsilon \propto \pi(\upsilon_{t+1}) := \frac{1}{\upsilon_{t+1}} \times \exp\left(-\frac{-2\rho D_{t+1}H_{t+1}+D_{t+1}^2}{2(1-\rho^2)} - \frac{D_{t+2}^2-2\rho D_{t+2}H_{t+2}+H_{t+2}^2}{2(1-\rho^2)}\right)$, for $0 < t+1 < T$. $D_{t+1} = \frac{\upsilon_{t+1}-\upsilon_t-\kappa(\theta-\upsilon_t)\Delta}{\sigma_v\sqrt{\upsilon_t\Delta}}$, and $H_{t+1} = \frac{p_{t+1}-p_t-\mu\Delta-J_{t+1}\xi_{t+1}}{\sqrt{\upsilon_t\Delta}}$.

Jump size related parameters for small jumps in $\mathcal{M}_4$ and $\mathcal{M}_5$ as well as all jumps in $\mathcal{M}_2$:

Posterior for μ_p: $\mu_p|\xi^y,\Theta\backslash\{\mu_p\} \sim \mathbb{N}\left(\frac{\mathcal{A}}{\mathcal{B}},\sqrt{\frac{1}{\mathcal{B}}}\right)$, where $\xi^y = \xi^{y+} + \xi^{y-}$, $\mathcal{A} = \sum_{t=0}^{T-1}\frac{\left(\xi_{t+1}^y\right)}{\sigma_p^2} + \frac{c^*}{C^*}$, $\mathcal{B} = \frac{T}{\sigma_p^2} + \frac{1}{C^*}$, $c^* = 0$ and $C^* = 100$ are hyperparameters of the prior of μ_p.

Posterior for σ_p^2: $\sigma_p^2|\xi^y,\Theta\backslash\{\sigma_p\} \sim \mathbb{IG}(\mathcal{A},\mathcal{B})$, where $\mathcal{A} = c^* + T$, $\mathcal{B} = C^* + \sum_{t=0}^{T-1}(\xi_{t+1}^y - \mu_p)^2$, $c^* = 10$ and $C^* = 40$ are hyperparameters of the prior of σ_p^2.

Jump size related parameters for LH jumps in $\mathcal{M}_4$ and $\mathcal{M}_5$ as well as all jumps in $\mathcal{M}_3$:

Posterior for μ_{pl^+}: $\mu_{pl^+}|\xi^{pl+},J^{pl^+},\Theta\backslash\{\mu_{pl^+}\} \sim \mathbb{IG}(\mathcal{A},\mathcal{B})$, where $\mathcal{A} = c^* + 2\sum_{t=0}^{T-1}\left(J_{t+1}^{pl^+}\right)$, $\mathcal{B} = C^* + 2\sum_{t=0}^{T-1}\left(\xi_{t+1}^{pl^+}J_{t+1}^{pl^+}\right)$ and $c^* = 10$ and $C^* = 20$ are hyperparameters of the prior of μ_{pl^+}. The values for μ_{pl^-} as well as μ_{y^+} and μ_{y^-} in model $\mathcal{M}_3$ can be estimated using the same method with variables in the respective dimension. Besides, if multi-layer price limits are applied, one can specify the jump size parameters for each price limit layer and estimate the additional parameters using the same method.

The jump size latent variables:

Posterior for ξ^y: $\xi_{t+1}^y|p,\upsilon,J_{t+1}^y = 1,\Theta \sim \mathbb{N}\left(\frac{\mathcal{A}}{\mathcal{B}},\sqrt{\frac{1}{\mathcal{B}}}\right)$, where $\mathcal{A} = \frac{H_{t+1}-\frac{\rho}{\sigma_v}D_{t+1}}{(1-\rho^2)\upsilon_t\Delta} + \frac{\mu_p}{\sigma_p^2}$, $\mathcal{B} = \frac{1}{(1-\rho^2)\upsilon_t\Delta} + \frac{1}{\sigma_p^2}$, $D_{t+1} = \upsilon_{t+1} - \upsilon_t - \kappa(\theta-\upsilon_t)\Delta$, and $H_{t+1} = p_{t+1} - p_t - \mu\Delta$. This is for the small jumps in $\mathcal{M}_4$ and $\mathcal{M}_5$ as well as all jumps in $\mathcal{M}_2$.

Posterior for ξ^{pl^+}: $\xi_{t+1}^{pl^+}|p,\upsilon,J_{t+1}^{pl^+} = 1,\Theta \sim \mathbb{N}\left(\frac{\mathcal{A}}{\mathcal{B}},\sqrt{\frac{1}{\mathcal{B}}}\right)\mathbf{1}_{0<\xi_{t+1}^{pl^+}<p_{t+1}-p_t}$, where $\mathcal{A} = \frac{D_{t+1}-\rho\sigma_v H_{t+1}}{(1-\rho^2)\sigma_v^2\upsilon_t\Delta} - \frac{1}{\mu_{pl^+}}$; $\mathcal{B} = \frac{1}{(1-\rho^2)\sigma_v^2\upsilon_t\Delta}$; $D_{t+1} = \upsilon_{t+1} - \upsilon_t - \kappa(\theta-\upsilon_t)\Delta$; and $H_{t+1} = p_{t+1} - p_t - \mu\Delta$. This is for the LH jumps in $\mathcal{M}_4$ and

$\mathcal{M}_5$ as well as all jumps in $\mathcal{M}_3$. The jump size variables for dimension pl^- as well as the positive and negative jumps in model $\mathcal{M}_3$ can be estimated using the same method with related parameters of the respective dimension.

Constant jump intensity in $\mathcal{M}_2$, $\mathcal{M}_3$ and $\mathcal{M}_4$:

Posterior for λ for small jumps in $\mathcal{M}_4$ and all jumps in $\mathcal{M}_2$: $\lambda|J^y \sim \mathbb{B}(\mathcal{A},\mathcal{B})$, where $J^y = J^{y+} + J^{y-}$, $\mathcal{A} = c^* + \sum_{t=0}^{T-1} J^y_{t+1}$, $\mathcal{B} = C^* + T - \sum_{t=0}^{T-1} J^y_{t+1}$, $c^* = 2$ and $C^* = 40$ are hyperparameters of the prior of λ. The jump intensity for jumps in $\mathcal{M}_3$ and LH jumps in $\mathcal{M}_4$ is measure separately for positive and negative jumps. The corresponding values can be estimated using the same method with J in the respective dimension.

Hawkes parameters in $\mathcal{M}_5$:

Posterior for $\boldsymbol{\lambda_0}, \boldsymbol{\alpha}, \boldsymbol{\beta}$: $p(\boldsymbol{\lambda_0}, \boldsymbol{\alpha}, \boldsymbol{\beta}|\boldsymbol{J_t}) \propto p(\boldsymbol{J_t}|\boldsymbol{\lambda_0}, \boldsymbol{\alpha}, \boldsymbol{\beta})p(\boldsymbol{\lambda_0}, \boldsymbol{\alpha}, \boldsymbol{\beta})$ and $p(\boldsymbol{J_t}|\boldsymbol{\lambda_0}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \left(\prod_{i=1}^{N} \lambda_{t_i}\right) exp\left(-\sum_{t=1}^{T} \lambda_t\right)$, where t_i denotes the times when $J_t = 1$. The posteriors of Hawkes parameters are sampled using the Metropolis-Hasting method. See more details of Bayesian estimation of Hawkes process in Rasmussen (2013).

Appendix D. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jbankfin.2024.107184.

References

Aït-Sahalia, Y., Cacho-Diaz, J., Laeven, R.J., 2015. Modeling financial contagion using mutually exciting jump processes. J. Financ. Econ. 117, 585–606.

Andersen, T.G., Benzoni, L., Lund, J., 2002. An empirical investigation of continuous-time equity return models. J. Finance 57, 1239–1284.

Baker, M., Wurgler, J., 2006. Investor sentiment and the cross-section of stock returns. J. Finance 61, 1645–1680.

Baker, M., Wurgler, J., 2007. Investor sentiment in the stock market. J. Econ. Perspect. 21, 129–152.

Bali, T.G., Cakici, N., Whitelaw, R.F., 2011. Maxing out: stocks as lotteries and the cross-section of expected returns. J. Financ. Econ. 99, 427–446.

Barber, B.M., Odean, T., 2008. All that glitters: the effect of attention and news on the buying behavior of individual and institutional investors. Rev. Financ. Stud. 21, 785–818.

Bates, D.S., 1996. Jumps and stochastic volatility: Exchange rate processes implicit in deutsche mark options. Rev. Financ. Stud. 9, 69–107.

Bates, D.S., 2000. Post-'87 crash fears in the S&P 500 futures option market. J. Econom. 94, 181–238.

Chen, J., Clements, M.P., Urquhart, A., 2023. Modeling price and variance jump clustering using the marked Hawkes process. J. Financ. Econom., nbad007.

Chen, J., Tang, G., Yao, J., Zhou, G., 2022. Investor attention and stock returns. J. Financ. Quant. Anal. 57, 455–484.

Chen, T., Dong, H., Lin, C., 2020. Institutional shareholders and corporate social responsibility. J. Financ. Econ. 135, 483–504.

Chen, T., Gao, Z., He, J., Jiang, W., Xiong, W., 2019. Daily price limits and destructive market behavior. J. Econom. 208, 249–264.

Clarke, J., Chen, H., Du, D., Hu, Y.J., 2020. Fake news, investor attention, and market reaction. Inf. Syst. Res. 32, 35–52.

Deb, S.S., Kalev, P.S., Marisetty, V.B., 2010. Are price limits really bad for equity markets? J. Bank. Finance 34, 2462–2471.

Du, D., Luo, D., 2019. The pricing of jump propagation: evidence from spot and options markets. Manag. Sci. 65, 2360–2387.

Duan, J.-C., Zhang, W., 2014. Forward-looking market risk premium. Manag. Sci. 60, 521–538.

Duffie, D., Pan, J., Singleton, K., 2000. Transform analysis and asset pricing for affine jump-diffusions. Econometrica 68, 1343–1376.

Engle, R.F., Ng, V.K., 1993. Measuring and testing the impact of news on volatility. J. Finance 48, 1749–1778.

Eraker, B., 2004. Do stock prices and volatility jump? Reconciling evidence from spot and option prices. J. Finance 59, 1367–1403.

Eraker, B., Johannes, M., Polson, N., 2003. The impact of jumps in volatility and returns. J. Finance 58, 1269–1300.

Fama, E.F., 1988. Perspectives on October 1987, or, what did we learn from the crash? Center for Research in Security Prices, Graduate School of Business.

Ferreira, M.A., Santa-Clara, P., 2011. Forecasting stock market returns: the sum of the parts is more than the whole. J. Financ. Econ. 100, 514–537.

Hawkes, A.G., 1971a. Point spectra of some mutually exciting point processes. J. R. Stat. Soc. B 33, 438–443.

Hawkes, A.G., 1971b. Spectra of some self-exciting and mutually exciting point processes. Biometrika 58, 83–90.

Hayo, B., Neuenkirch, M., 2015. Self-monitoring or reliance on media reporting: how do financial market participants process central bank news? J. Bank. Finance 59, 27–37.

Heston, S.L., 1993. A closed-form solution for options with stochastic volatility with applications to bond and currency options. Rev. Financ. Stud. 6, 327–343.

Hirshleifer, D., Hsu, P.-H., Li, D., 2013. Innovative efficiency and stock returns. J. Financ. Econ. 107, 632–654.

Jeon, Y., McCurdy, T.H., Zhao, X., 2022. News as sources of jumps in stock returns: evidence from 21 million news articles for 9000 companies. J. Financ. Econ. 145, 1–17.

Jiang, F., Lee, J., Martin, X., Zhou, G., 2019. Manager sentiment and stock returns. J. Financ. Econ. 132, 126–149.

Jiang, L., Liu, J., Peng, L., Wang, B., 2022. Investor attention and asset pricing anomalies. Rev. Finance 26, 563–593.

John, K., Li, J., 2021. Covid-19, volatility dynamics, and sentiment trading. J. Bank. Finance 133, 106162.

Kadilli, A., 2015. Predictability of stock returns of financial companies and the role of investor sentiment: a multi-country analysis. J. Financ. Stab. 21, 26–45.

Kim, K.A., Rhee, S.G., 1997. Price limit performance: evidence from the Tokyo stock exchange. J. Finance 52, 885–901.

Kou, S., Yu, C., Zhong, H., 2017. Jumps in equity index returns before and during the recent financial crisis: a Bayesian analysis. Manag. Sci. 63, 988–1010.

Kou, S.G., Wang, H., 2004. Option pricing under a double exponential jump diffusion model. Manag. Sci. 50, 1178–1192.

Lauterbach, B., Ben-Zion, U., 1993. Stock market crashes and the performance of circuit breakers: empirical evidence. J. Finance 48, 1909–1925.

Lazar, E., Qi, S., 2022. Model risk in the over-the-counter market. Eur. J. Oper. Res. 298, 769–784.

Lehmann, B.N., 1989. Commentary: volatility, price resolution, and the effectiveness of price limits. In: Regulatory Reform of Stock and Futures Markets. Springer, pp. 107–111.

Leippold, M., Wang, Q., Zhou, W., 2022. Machine learning in the Chinese stock market. J. Financ. Econ. 145, 64–82.

Li, H., Wells, M.T., Yu, C.L., 2008. A Bayesian analysis of return dynamics with Lévy jumps. Rev. Financ. Stud. 21, 2345–2378.

Li, N., Richardson, S., Tuna, İ., 2014. Macro to micro: country exposures, firm fundamentals and stock returns. J. Account. Econ. 58, 1–20.

Lin, F., Qiu, Z., Zheng, W., 2023. Cranes among chickens: the general-attention-grabbing effect of daily price limits in China's stock market. J. Bank. Finance 150, 106818.

Liu, J., Thomas, J., 2000. Stock returns and accounting earnings. J. Account. Res. 38, 71–101.

Maneesoonthorn, W., Forbes, C.S., Martin, G.M., 2017. Inference on self-exciting jumps in prices and volatility using high-frequency measures. J. Appl. Econom. 32, 504–532.

Merton, R.C., 1976. Option pricing when underlying stock returns are discontinuous. J. Financ. Econ. 3, 125–144.

Neumann, M., Prokopczuk, M., Simen, C.W., 2016. Jump and variance risk premia in the S&P 500. J. Bank. Finance 69, 72–83.

Rasmussen, J.G., 2013. Bayesian inference for Hawkes processes. Methodol. Comput. Appl. Probab. 15, 623–642.

Santa-Clara, P., Yan, S., 2010. Crashes, volatility, and the equity premium: lessons from S&P 500 options. Rev. Econ. Stat. 92, 435–451.

Seasholes, M.S., Wu, G., 2007. Predictable behavior, profits, and attention. J. Empir. Finance 14, 590–610.

Yi, Z., Mao, N., 2009. Research on investor sentiment measurement in Chinese stock market: construction of CICSI (translated from mandarin). Financ. Res. 11, 174–184.